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COUNTRY REPORT SCOPING STUDY CLIMATE SMART RICE THAILAND

**Promoting Global Best Practices and
Scaling of Low Emissions Technologies by
Engaging the Private and Public Sectors in the
Paddy Rice Sector**

June 2021



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ACRONYMS

ASEAN	Association of Southeast Asian Nations
AWD	Alternate Wetting and Drying
BAAC	Thailand Bank for Agriculture and Agricultural Cooperatives
BMZ	German Federal Ministry for Economic Cooperation and Development
BUR	Biennial Update Report
CCAC	Climate and Clean Air Coalition
CO ₂ eq.	Carbon dioxide equivalent
CRC	Community Rice Centre
GAP	Good Agriculture Practices
GEF	Global Environment Facility
GHG	Greenhouse Gas
GIZ	Deutsche Gesellschaft für Internationale Zusammenarbeit
GWP	Global Warming Potential
ha	hectare
ICT	Information and Communication Technology
IPCC	Intergovernmental Panel on Climate Change
IPPU	Industrial Processes and Product Use
IRRI	International Rice Research Institute
LLL	Laser Land Levelling
M&E	Monitoring and Evaluation
MoAC	Thailand Ministry of Agriculture and Cooperatives
MoNRE	Thailand Ministry of Natural Resources and Environment
MRV	Measurement, Reporting and Verification
MSVC	Market Oriented Smallholder Value Chains
Mt	Mega tonne
NAMA	Nationally Appropriate Mitigation Action
NCCC	National Committee on Climate Change Policy
NDC	National Determined Contribution
NGO	Non-Governmental Organisation
OAE	Office of Agricultural Economics
ONEP	Office of Natural Resources and Environmental Policy and Planning
PCD	Pollution Control Department

PI	Performance Indicators
PM	Particulate Matter
SARI	Sustainable Aromatic Rice Initiative
SDGS	Sustainable Development Goals
SLCP	short-lived climate pollutants
SRP	Sustainable Rice Platform
SSNM	Site Specific Nutrient Management
TBUR	Third Biennial Update Report
TEEB	The Economics of Ecosystem and Biodiversity
TGCP	Thai-German Climate Programme
TRD	Thai Rice Department
UNEP	United Nations Environment Programme
UNFCCC	United Nations Framework Convention on Climate Change

EXECUTIVE SUMMARY

This report provides an overview of current national and international initiatives around low emissions rice in Thailand. It also assesses the political, institutional, and national capacity and readiness to absorb large-scale funding to support mitigation measures in the rice sector under the Climate and Clean Air Coalition (CCAC) initiative that is led by the United Nations Environment Programme (UNEP). This report is a deliverable of the CCAC project, “Promoting Global Best Practices and Scaling of Low Emissions Technologies by Engaging the Private and Public Sectors in the Paddy Rice Sector” implemented by the UNEP Regional Office of Asia and the Pacific. This project will identify opportunities and approaches to mitigate the Short-Lived Climate Pollutants (SLCPs) from rice sector by fostering sustainable, climate-smart practices in rice cultivation management at farmers level, and to improve the capacity of key agencies to scale-up the SLCP mitigation packages.

Rice is a staple food for more than half of the world's population and forms an essential part of the diet in most Asian countries. It also produces greenhouse gas emissions including methane. Methane is produced during anaerobic decay of organic material in continuously flooded rice paddies. Paddy rice responsible for 8% of global anthropogenic methane emissions each year and is also a source of nitrous oxide emissions from fertilizers. Methane emissions from rice production were 26.6 Mt CO₂eq. per year for Thailand in 2016, 51% of the country's agricultural emissions. The rice sector could mitigate as much as 48% of global methane emissions from paddy rice through wide scale adoption of water management practices such as alternate wetting and drying (AWD). However, irrigation manage-

ment is frequently cited without specifics on how this practice will be scaled up across the rice sector. There is a need for greater clarity in technology assessment, impact monitoring, and capacity building in this area.

Paddy rice is a major crop in Thailand, economically important for rural employment and for domestic and export consumption. The Northeast, North, and Central regions produce most Thailand's rice. About three-quarters of rice in Thailand is rainfed and one-quarter is irrigated. Irrigation is mostly located in central region, where rice can be grown two to three times per year. As irrigated rice produces three times the methane of rainfed rice, this is the main area which could benefit from AWD (which requires irrigation). Lack of controlled irrigation is a barrier to productivity and AWD adoption. Climate change also impacts farmers in Thailand, as increased temperatures contribute to both drought and flood events.

Around 4.2 million farming households relied on rice cultivation in 2019. Agriculture employed 31% of Thailand's labour force with 34% of men and 28% of women in Thailand. The rural workforce in Thailand is aging and the number of farmers is declining as younger people move to urban areas. The average age of Thai rice farmers is 50 years old and due to the aging rural population, investment in new agricultural technologies has declined since 2000. On the other hand, farmers in Thailand have good access to credit. More than 90% of farm households have access to the bank's credit. Around 40% of Thai farming households were living below the poverty line in 2018.

A range of ongoing initiatives aim to improve the Thai rice sector. These range from eco-

nomic support of farmers, to development of value chains, to promoting adoption of low-emission practices. The flagship international cooperation project related to climate is the Thai Rice NAMA (Nationally Appropriate Mitigation Action), which is funded by the NAMA Facility. This project promoted AWD and laser land levelling, as well as a financial incentive mechanisms and policy change.

Thailand's Third Biennial Update Report notes four areas in which support is needed: 1) improvement of technology development and transfer; 2) technical support / assistance to support the transfer of technical data, know-how and knowledge, including MRV and M&E System; 3) capacity building; and 4) financial support. The UNEP-CCAC paddy rice project aims to support all four areas through targeted interventions and working with other national and international partners in Thailand, especially the Sustainable Rice Platform (SRP). Application of the SRP Standard reduces methane mitigation in irrigated areas but also has co-benefits for reduction of nutrient and pesticide pollution and straw burning as well as supporting climate adaptation. There is a high potential for Information and Communication Technology tools to aid adoption of SRP Standard and support measurement, reporting, and verification (MRV) of paddy rice emissions. A variety of financing mechanisms are available to support this work, including: green loans; improved crop insurance; socially responsible

contract farming; multi-donor funds; carbon credits; and green bonds for sustainable rice.

The following principles should be applied to future work on paddy rice methane mitigation in Thailand:

- **It is important to follow an integrated approach for mitigation and adaptation. A pure mitigation focus (e.g., removal of Short-Lived Climate Pollutants) cannot achieve results.**
- **Follow a landscape approach (farming system approach). This entails that rice does not grow in isolation.**
- **Develop Nationally Appropriate Mitigation Activities and financing strategies for rice sector with easy-to-follow obligations and more accountability for stakeholders (certainly with adaptation co-benefits).**
- **Harness SRP's potential as an inclusive, comprehensive, and already tested entry point.**
- **The concept needs to be flexible. New openings may emerge to support intervention packages and must be included.**
- **It is essential to follow a multi-partner and multi-stakeholders' approach. Federal and regional ministries, the rice millers, technology developers, communication technology, media - have to be on board for lending their respective strengths.**

CHAPTER 1. INTRODUCTION

1.1. Background

This report provides an overview of current national and international initiatives around low emissions rice in Thailand. It also assesses the political, institutional, and national capacity and readiness to absorb large-scale funding to support mitigation measures in the rice sector under the Climate and Clean Air Coalition (CCAC) initiative that is led by the United Nations Environment Programme (UNEP). This report is a deliverable of the CCAC project, “Promoting Global Best Practices and Scaling of Low Emissions Technologies by Engaging the Private and Public Sectors in the Paddy Rice Sector” implemented by the UNEP Regional Office of Asia and the Pacific.

CCAC is a global coalition focused on reducing short-lived climate pollutants (SLCPs). SLCPs are powerful climate forcers, some of which are also air pollutants. SLCPs include methane, hydrofluorocarbons (HFCs), tropospheric ozone, and black carbon. CCAC promotes eleven initiatives that have great potential to reduce climate impacts. Four of the initiatives carry out work across sectors to accelerate emission reductions, and seven of the initiatives focus on single sector. In 2019 Thailand joined the CCAC to further develop assessments of air pollution and greenhouse gas emissions in the Asia Pacific region and to determine mitigation opportunities in the region. In 2021 Thailand and the CCAC began a new initiative under CCAC’s Supporting National Planning and Action to reduce SLCPs (SNAP) Initiative (CCAC, 2021). This new initiative identifies solutions to reduce emissions of SLCPs across Thailand. It is led by the Pollution Control Department (PCD), under the Ministry of Natural Resources and Environment (MoNRE). PCD is the CCAC focal point for Thailand.

Rice is a staple food for more than half of the world’s population and forms an essential part of the diet in most Asian countries. It also produces greenhouse gas emissions including methane (CH_4). Methane is produced during anaerobic decay of organic material in continuously flooded rice paddies. Rice responsible for 8% of global anthropogenic methane (UNEP, 2021) emissions each year and is also a source of nitrous oxide (N_2O) emissions from fertilizers. Methane emissions from rice production were 26.6 Mt CO_2eq per year for Thailand in 2016, 51% of the country’s agricultural emissions (ONEP, 2020). The rice sector could mitigate as much as 48% of global methane emissions from paddy rice through wide scale adoption of water management practices such as alternate wetting and drying (AWD). However, irrigation management is frequently cited without specifics on how this practice will be scaled up across the rice sector. There is a need for greater clarity in technology assessment, impact monitoring, and capacity building in this area.

To that end and to address other sustainability issues around rice cultivation, the Sustainable Rice Platform (SRP) has developed the world’s first voluntary standards for sustainable rice cultivation. These standards enable farmers to cultivate and manage rice production and value chain sustainably, conserving resources, and reducing climate impacts while also improving their livelihoods. The SRP is a global multi stakeholder alliance of over 100 institutional members from public, private, research, civil society, and the financial sector. It was originally co con-

vened by the International Rice Research Institute (IRRI) and the United Nations Environment Programme (UNEP), and later Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH. SRP is now an independent member association with over 100 members. As a multi-stakeholder platform, SRP works with its members and partners to transform the global rice sector to be more sustainable by improving smallholder livelihoods, reducing the social, environmental, and climate footprint of rice production. The SRP is offering the global rice market an assured supply of sustainably produced rice by developing and providing a sustainable rice cultivation standard, performance indicator (PI), and assurance framework to support the rice value chain's actors in transforming the rice sector to be more sustainable.

The UNEP-CCAC project, “Promoting Global Best Practices and Scaling of Low Emissions Technologies by Engaging the Private and Public Sectors in the Paddy Rice Sector” draws on the SRP initiative to support Thailand to reach its NDCs (National Determined Contributions) target. This project will identify opportunities and approaches to mitigate the SLCPs from rice sector by fostering sustainable, climate-smart practices in rice cultivation management at farmers level, and to improve the capacity of key agencies to scale-up the SLCP mitigation packages. To reach a sustainable solution, a joint effort of various multi-stakeholders in the Thai rice value chain is essential. The project will synthesize lessons learned and foster cooperation with other rice projects in Thailand—such as the Thai Rice NAMA (Nationally Appropriate Mitigation Action), Market Oriented Smallholder Value Chains (MSVC), and Thai-German Climate Programme (TGCP)—projects currently led by GIZ. The project will also engage other SRP members in Thailand under the SRP National Chapter for Thailand.

To achieve large-scale mitigation of SLCP emissions from paddy rice cultivation across Thailand's key rice-growing areas, the project in its first phase conducted a scoping study to assess the feasibility of SLCP mitigation activities within the national context. The results of that scoping study are reported here. A national consultant assessed the feasibility of scaling up low emissions rice practices in Thailand in two parts: 1) the technical potential of packages of SLCP mitigation measures; 2) a socio-economic assessment of rice mitigation options in Thailand. Based on lessons learned from ongoing initiatives, promising technologies and approaches will be combined into a package of SLCP ‘best-practice’ recommendations, that will also consider potential synergies and trade-offs in implementation of new technology packages at local level.

1.2. Objectives of the study

This study identifies potential SLCP mitigation packages relevant to the national context in Thailand and assesses the socio-economic context and potential impact of the mitigation packages. This study was done in collaboration with national and international partners, experts, and stakeholders. The main components of the project are:

1. Prepare a report summarizing SLCP mitigation packages for priority rice areas identified for Thailand.
2. Assess the socio-economic impacts of most promising SLCP mitigation packages in key rice growing environments.
3. Consult the draft mitigation packages / detailed country reports describing technical and economic feasibility of mitigation options with the local stakeholders as well as UNEP.

1.3 Methodology

This study used a qualitative research method to collect data from both primary and secondary sources. The methodology followed the steps described below:

1. Preliminary research

The preliminary research focused on secondary data collection from the following sources: journals, national policy, other projects reports, development initiatives, previous research, and other publications. Information gathered from various stakeholders were analysed based on the stakeholder functions reflected in the sustainable rice value chain mapping in Thailand. The list of the stakeholders is reflected in Annex 1. The objective of this step was to gain insights into the current status of mitigation efforts in Thailand, particularly with reference to the rice sector.

2. Stakeholder Consultations

To obtain the required data on multiple topics, expert interviews and surveys were conducted with key stakeholders involved in Thailand's rice value chain, selected based on stakeholder mapping. This step aimed to help build an understanding of the condition and challenges to adoption of currently available mitigation measures, and to identify potential improvements or scale-up strategies. (See Annex 1 for list of participating stakeholder representatives.)

Fifteen stakeholder representatives were interviewed from December 2020 to February 2021. All interviews were conducted online due to the COVID-19 pandemic. Respondents were selected based on their experiences in implementing or supervising climate change or sustainable rice production projects in Thailand. Respondents comprised of 6 females and 9 males. Due to the limitations imposed by language and the pandemic, interviews with farmers or farmer representatives could not be arranged under this study.

3. Data Analysis

To achieve the study objective, interview data were analysed using a Qualitative Content Analysis method. This method is widely used in qualitative research especially in social research. This method was done by working through several journals and text passages, interviews and open questions then categorised the data into perceived situations. There is no perfect and standard design study with this method (Bengtsson. 2016). The researcher's self-reflection is an essential part in this method (Burnard, 1995). The assessments were conducted from a socio-economic and technical perspective, which were then compiled into the current country report.

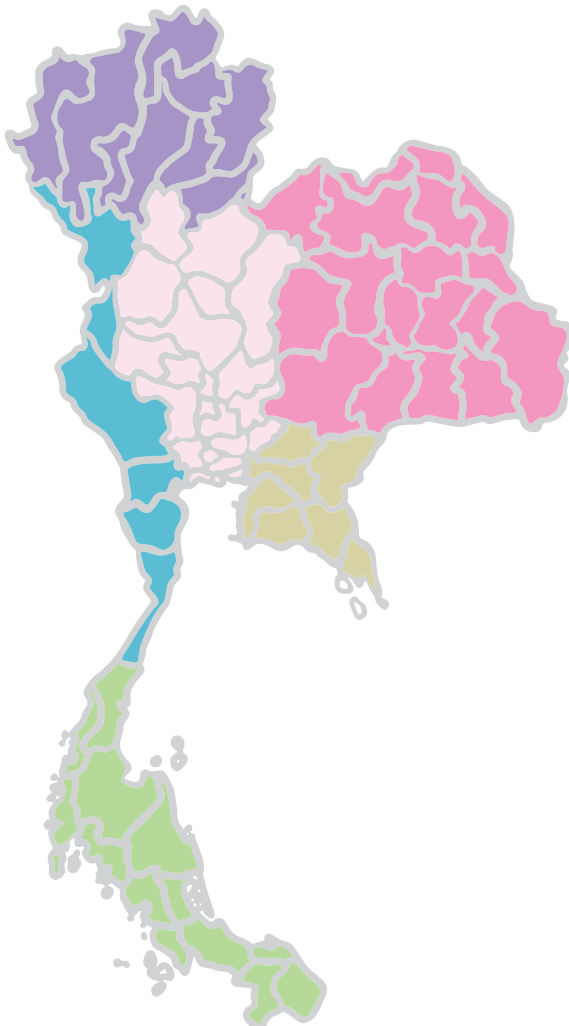
CHAPTER

2.

THEORETICAL BACKGROUND

The Kingdom of Thailand is geographically located in Southeast Asia with an area of 513,115 km² with a total coastline of about 2,815 km. The total population as of 2018 was 66,413,979 inhabitants, making Thailand the 20th most populated country in the world. The Thai population is projected to gradually increase until 2030 and then may decline from that point. Experts predict that in 2030 the population would reach a peak of 67,140,000 inhabitants. The birth rate would decline with a concurrent increase in life expectancy, which has shaped Thailand's population into an "aging society." It is projected that in 2040 the elder population group of 60 years and older would dominate the population over the "productive" population group (between 15 to 59 years old) and more than double of the population group of 0 to 14 years old. The country should anticipate shrinking labour force which could harm Thailand's economy through a decrease in labour productivity (ONEP, 2020).

Figure 1. Thailand's Six Regions Division



Thailand is administered into 6 official regions: 1) North, 2) West, 3) East, 4) Northeast, 5) South, and 6) Central, and divided into 76 provinces – not including Bangkok (the capital city) (Figure 1). Thailand has three tiers of administration: central, provincial, and district administrations. The central administration is on the national level, such as the ministries, bureaus, and departments. Bangkok as a capital has special administrative area, locally governed by the Bangkok Metropolitan Administration.

Thailand's climate varies considerably across the six regions and is influenced primarily by the Southwest and Northeast monsoons. The Southwest monsoon starts in May and brings a stream of warm, moist air from the Indian Ocean towards Thailand, causing abundant rain across the country. The northeast monsoon starts in October, bringing cold, dry air from the anticyclone over the Chinese mainland. These two seasonal monsoons result in three seasons: Rainy season (Southwest monsoon season) from mid-May to mid-October, Winter (Northeast monsoon season) from mid-October to mid-February, and lastly the Summer (pre-monsoon season) from mid-February to mid-May. The rainy season is the most important season for the agriculture sector and of course for rice cultivation. Farmers with access to reliable irrigation are also able to grow rice following the end of the rainy season, but this area is much smaller than rainy-season production (ONEP, 2020).

This chapter will next discuss climate change in Thailand. This includes how the Thai government is preparing the country through its national strategy and how climate change affects the country's rice sector.

2.1. Climate Change in Thailand

On October 20th, 2020, Thailand updated its nationally determined contribution (NDC) to the United Nations Framework Convention on Climate Change (UNFCCC). Initially, Thailand intended to reduce its greenhouse gas emissions by 20 percent from the projected business-as-usual (BAU) level by 2030. In the updated document submitted in 2020, Thailand stated that this target could be increased to 25% with technology, financial, and capacity building support. Thailand produced 241 Mt (mega tonnes) of CO₂ in 2018 which represents a 0.85% share of total global GHG emissions (ONEP, 2020; IEA, 2019). The energy—which covers electricity and heat producers, transport, and industry—is the largest source of Thailand's GHG emissions. In the energy and transportation sectors Thailand has successfully reduced 58. MtCO₂eq (carbon dioxide equivalent) by 2018, a reduction of 16 percent from the BAU (ONEP, 2020).

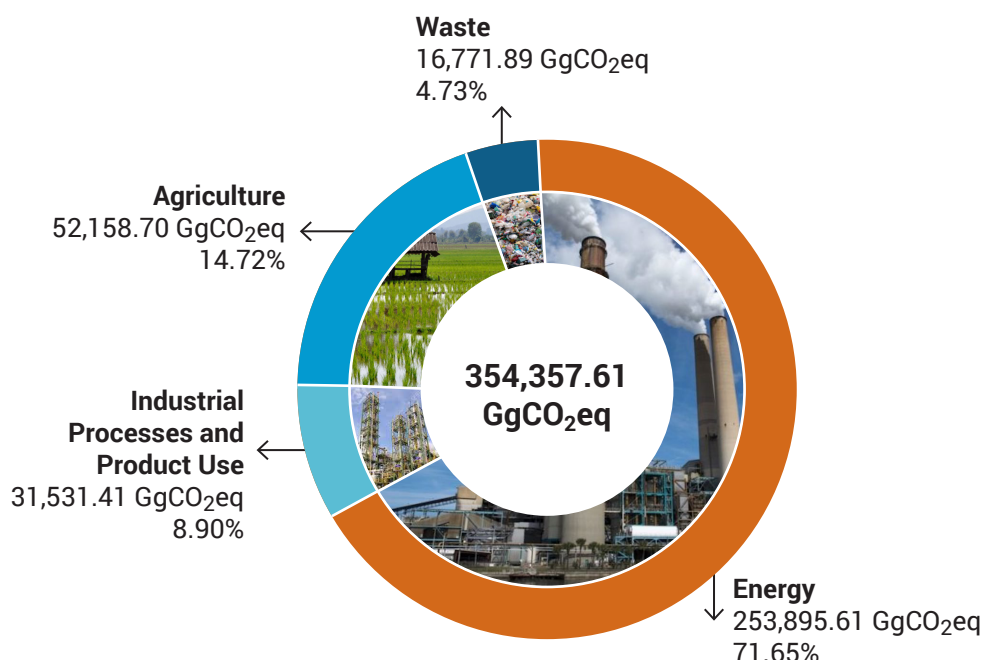
As of 2016, Thailand's total direct emissions (excluding those from Land Use and Land-Use Change and Forestry (LULUCF) sectors) were 354 MtCO₂eq, while net removals of CO₂ amounted to 91 Mt CO₂eq. Net GHG emissions were therefore 263 MtCO₂eq, broken down by sector as follows:

1)	Energy:	254 MtCO ₂ eq
2)	Agriculture:	54 MtCO ₂ eq
3)	Industrial Processes and Product Use (IPPU):	32 MtCO ₂ eq
4)	Waste:	17 MtCO ₂ eq

The proportion of GHG emissions in the energy sector accounted for 72% of total emission sources in 2016, followed by the agriculture, IPPU and waste sectors (Figure 2). Overall, Thailand's GHG emissions represent less 1% of global emissions. Despite this, Thailand is nevertheless ranked the world's 8th most affected by climate change under the Global Climate Risk Index, over the period 1999 to 2018. The increasing severity of impacts prompted Thailand to elevate the issue as a national priority. Thailand has since established inter-ministerial commit-

tees to oversee climate policy development, implementation, and a comprehensive national system to monitor, evaluate, and report on progress.

Figure 2. Total GHG Emissions by sector as of 2016 (excluding LULUCF)



The National Committee on Climate Change Policy (NCCC) was established by the Thai government in 2007, is chaired directly by the Prime Minister, and has members from both public and private sectors. The NCCC is mandated to define national climate policies and establish guidelines, systems, and mechanisms aligned with international conventions and protocols on climate change. NCCC comprises the following five sub-committees:

- 1) Climate Change Policy and Planning Integration,
- 2) Climate Change Knowledge and Database,
- 3) Climate Change Negotiation and International Cooperation,
- 4) Public Relations and Actions for Climate Empowerment,
- 5) Climate Law.

In addition to the Chairperson, two vice-chairpersons—represented by the respective Ministers—provide support from the two key ministries: Ministry of Natural Resources and Environment (MoNRE) and the Ministry of Foreign Affairs (MoFA) (ONEP, 2020).

Climate change is a major priority under Thailand's National Strategic Plan (2018 – 2037) for a long-term approach. As reflected in its Third Biennial Update Report (TBUR) to UNFCCC, Thailand will implement its climate targets through the NDC Roadmap on Mitigation 2021 – 2030, the NDC Sectoral Action Plans, and the NDC Supportive Action Plan. Development of these strategies was undertaken in close consultation with key stakeholders, including several public consultations at both the national and local levels (ONEP, 2020). The NDC Sectoral Action Plan identifies achievable emission reduction targets. The Plan focuses on Energy, Transportation, IPPU, and Waste Management, which are further subdivided into 15 groups of mitigation measures. No explicit mitigation target has been set for agriculture.

Climate change adaptation and mitigation are also included in other social and economic development plans in Thailand. The Thai government recognizes the opportunity presented by recovery plans to include ecosystem management, climate resilience, and sustainable

development as part of its 'Build Back Better' plans, including through implementation of NDC (ONEP, 2020). With regards creating an inclusive and sustainable economy, the Thai government also launched the Thailand 4.0 strategy with a commitment to the Sustainable Development Goals (SDGs), linked to the philosophy of the "Sufficiency Economy," a concept developed by His Majesty King Bhumibol Adulyadej of Thailand.¹ The target is to achieve "Security, Prosperity and Sustainability," to create a "value-based economy" (FAO, 2018). The Thailand 4.0 strategy aims to transform traditional farming to "smart farming."²

2.2 Measurement, Reporting and Verification (MRV) system for climate change

Measurement, Reporting, and Verification (MRV) is a system that promotes the transparency of greenhouse gas emissions and mitigation actions. Non-Annex I countries belonging to the UNFCCC should follow the MRV guidelines laid out by the UNFCCC. According to the *Handbook on Measurement, Reporting and Verification for Developing Country Parties* (UNFCCC, 2014), **measurement** occurs at the national level and refers to the measurement of GHG emissions and the national GHG inventories. **Reporting** is done through the submission of National Communications (NCs) and Biennial Update Reports (BURs). NCs are to be submitted every four years, and BURs are to be submitted every two years. **Verification** is addressed at the international level through International Consultation and Analysis (ICA) of BURs, which is a process to increase the transparency of mitigation actions and their effects, and support needed and received. Provisions for verification at the national level that are part of the national MRV framework are to be reported in the BURs (UNFCCC, 2014). Soon the Enhanced Transparency Framework will become to MRV mechanism for the UNFCCC. Countries will report their GHG inventory and progress towards their NDCs in a biennial transparency report.³

Countries report their GHG emissions in a GHG inventory report, which is guided by IPCC methodologies. Under the 2nd NCCC Sub-committee on Climate Change Knowledge and Database, Thailand created five sectoral Working Groups, which review the GHG inventory, NCs, and BURs that are submitted to UNFCCC. The Working Groups are organized by Office of Natural Resources and Environmental Policy and Planning (ONEP), with the lead agencies from the five sectors:

1. **Energy:** led by the Energy Policy and Planning Office and the Office of Transport and Traffic Policy and Planning
2. **IPPU:** led by the Department of Industrial Works
3. **Agriculture:** led by the Office of Agricultural Economics
4. **LULUCF:** led by Department of National Parks, Wildlife and Plant Conservation
5. **Waste:** led by the Pollution Control Department.

To support the national inventory system, Thailand is currently developing an IT tool called the Thailand GHG Emissions Inventory System (TGEIS) to store emissions data and estimates for the country. This tool is based on the 2006 IPCC Guidelines and will facilitate the work of various agencies to collect inventory report for each economic sector, control data quality, and integrity and serve as a platform for information exchange among relevant agencies throughout the inventory process (TGCP, 2019). The latest reporting of Thailand was submitted in December 2020, via the Third Biennial Update Report (TBUR) to UNFCCC. A fourth BUR is under development.

¹For more information: <http://thaiembassy.se/en/monarchy/philosophy-of-sufficiency-economy/>

²For more information: <https://thaiembdc.org/thailand-4-0-2/>

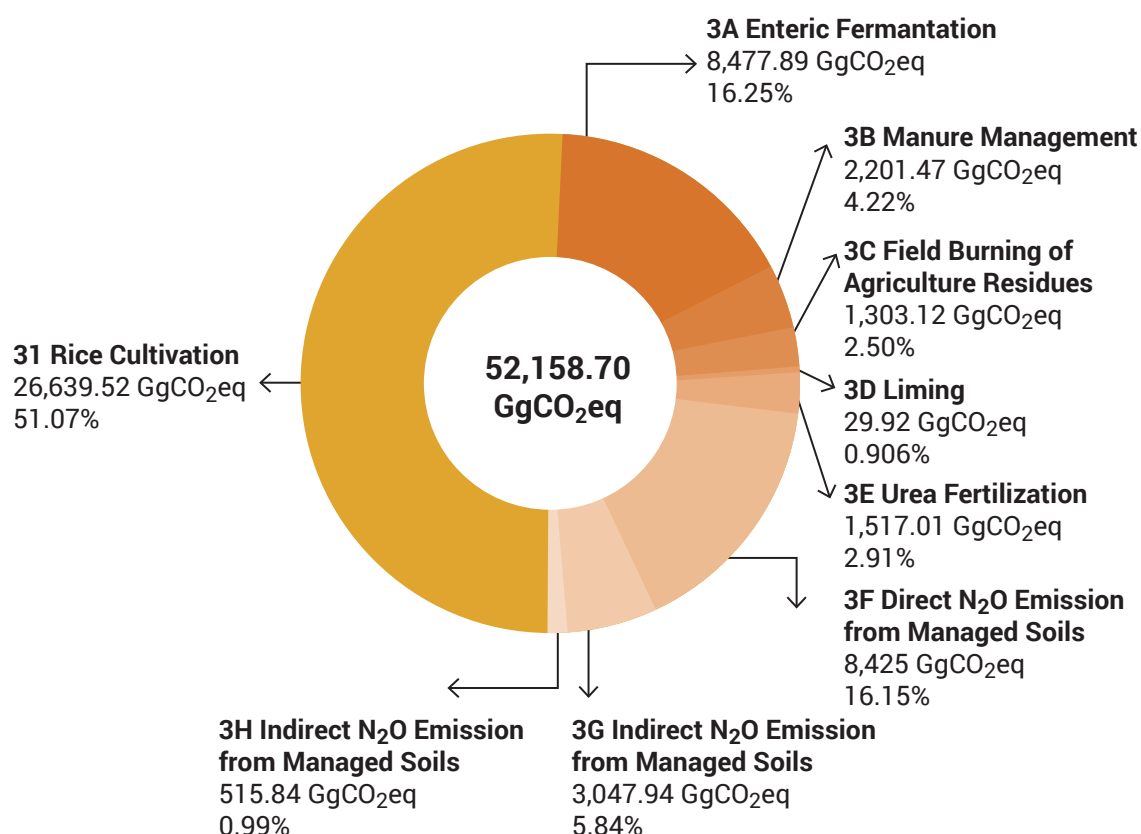
³<https://unfccc.int/process-and-meetings/transparency-and-reporting/reporting-and-review-under-the-paris-agreement>

Agriculture sector data is nevertheless required for GHG inventory reports. For agriculture, the Office of Agricultural Economics (OAE) is the lead agency within MoAC mandated to collect the relevant data. Data for the GHG inventory are derived from rice cultivation, livestock farming, agricultural soils, field burning, and open burning of agricultural residues (ONEP, 2020). On the MRV – inventory data collection, ONEP identified several areas of support that are needed in the agriculture sector, mainly on the 1) MRV data calculation. More accurate GHG estimation is needed, e.g., to refine the relevant emission factors. Moreover, due to the issue of open burning, more appropriate methods are needed, including use of satellite imagery. Finally, given the numerous challenges in gathering data, collaboration with the private sector would be needed. 2) Secondly, QA / QC (Quality Assurance / Quality Control) of the data needs to be strengthened to ensure data integrity and alignment with IPCC criteria. Further capacity building on the 2006 IPCC Guideline is needed to support government officers who collect and analyse the data (ONEP, 2020). ONEP has also published an MRV Manual for Agriculture Sector in 2017.⁴

2.3. Overview of Agricultural Greenhouse Gas Emissions in Thailand

The agriculture sector is the 2nd biggest contributor to the country's total GHG emissions, accounting for 52 MtCO₂eq in 2016. From this total, rice cultivation contributed the largest share (51%) at 27 MtCO₂eq (Figure 3), followed by the combination of direct and indirect nitrous oxide emissions from managed soils (22%, 11 MtCO₂eq). Field burning of agricultural residues contributed 2.5% (1.3 MtCO₂eq) of total agricultural emissions. These estimates are from the Third Biennial Update Report prepared by ONEP, which are regularly updated.

Figure 3. GHG emissions in Agriculture sector 2016.



Source: Thailand's Third Biennial Update Report (ONEP 2020).

⁴<https://www.asean-agrifood.org/thai-government-measures-local-greenhouse-gas-emissions/>

2.4. Rice Sector in Thailand

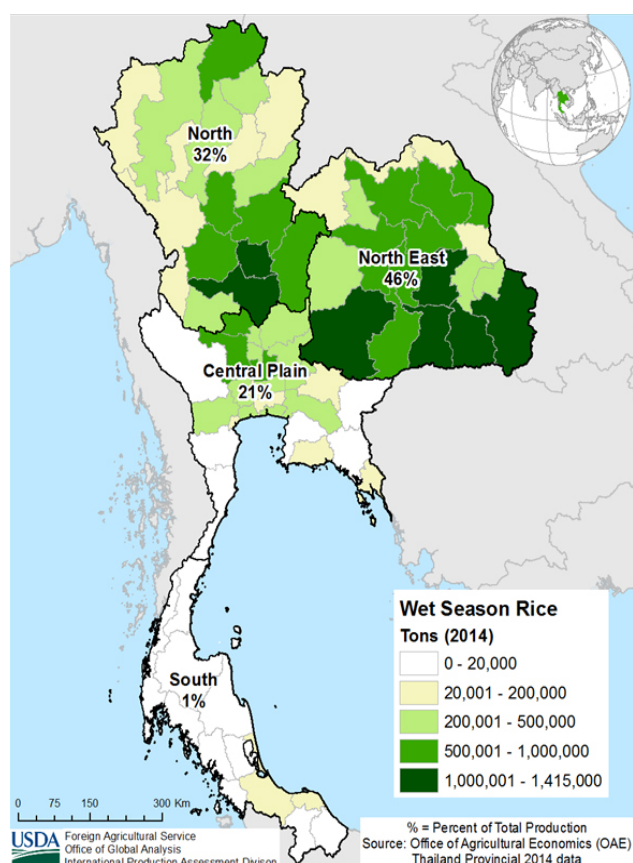
Rice is grown in Thailand mostly during the wet season (Figure 4) but also in the dry season (Figure 5) when irrigation is possible. As of 2020–21, Thailand's rice production area was 10 million hectares (ha). In dry season 2020–21, only 1 million ha rice area is cultivated due to water limitation. At most around 2 million ha rice can be grown during dry season depending on water availability. Thailand produced nearly 28 million tonnes of paddy rice (composed of various rice varieties) in the wet season 2020–21 (OAE, 2021). Geographically rice is grown throughout the country, but the most important rice-growing region can be found in the Northeast, North, and Central of Thailand with 49%, 25%, and 22% respectively, of the country's total rice growing area (Liese et al., 2014). Historically central Thailand was the major producer of rice, but in the last decade the north and the north-eastern part of Thailand have increased their rice production. The north and north-eastern areas produce about 80% of Thailand's Jasmine rice and other high-value varieties for domestic use and export.

Rice in Thailand is planted in two different irrigation regimes: the rain-fed and irrigated system. Nearly 74% of rice by area is rain-fed, which commonly practiced in the upland regions of Northeast and most of the North of Thailand. Only 10% of rice in the northeast is irrigated. Water scarcity has hampered the development of irrigation systems, particularly in the Northeast region, whereas in the Central and Lower Northern area, irrigation systems are more widespread and advanced.

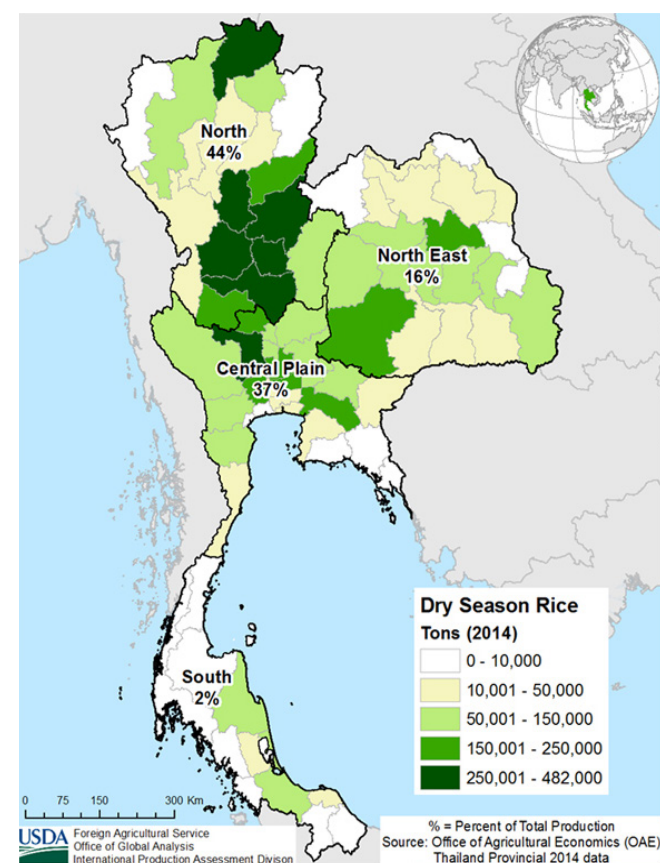
Figure 4. Wet season rice production in Thailand, 2014.

Figure 5. Dry season rice production in Thailand, 2014.

THAILAND: Wet Season Rice Production



THAILAND: Dry Season Rice Production



Availability of irrigation is a key determinant of the different patterns of rice cultivation in the Central and Northeast regions. In the largely rain-fed Northeast region, only a single crop per year can be produced. Predominantly, traditional varieties are favoured, particularly

fragrant Jasmine (Hom Mali) or KDML 105. Although the quality and price are higher for fragrant rice, yields are lower at only around 2 tonnes per ha. In contrast, in the irrigated systems prevailing in the Central plains, modern rice varieties are generally grown, with at least two crops per year, or five crops within two years. Average yields are also much higher at around 4 tonnes per ha under these more intensive production conditions (Liese et al., 2014; Jondée et al., 2006). Thailand's rice productivity lags behind other countries in the region due to the lack of controlled irrigation, the prevalence of high value but lower yielding Hom Mali, and its demographic characteristics.

Table 1 below summarizes the different characteristics of typical rain fed and irrigated rice farms in the two study regions based on a study by Agribenchmark in the Northeast and Central regions.

Table 1: Characteristic of the typical rain-fed and irrigated rice farm in Thailand

	Rain-Fed Farm <i>1 crop cycle / year</i>	Irrigated Farm <i>2 crop cycles / year</i>
Farm size (ha)	5	4.3
Soil type	Sandy	Loamy
Yield (ton/ha/year)	2.86	5.6
Market Price (un-milled paddy, THB/ ton)	20,000	12,500
Varieties	KDML 105, RD15	RD47 and Pathumthani 1
Crop Establishment	Dry direct seeding	Pre-germinated direct seeding
Land Preparation	- 1 st ploughing with hired 4-wheel tractor - 2 nd ploughing with owned power tiller	- 1 st ploughing with hired small tractor - Ploughing with owned power tiller
Seed	Saved seed	Bought seed
Fertilizer	Compound fertilizer (N-P-K) - Before broadcasting: 16-6-8 - 40 days after broadcasting: 46-0-0 - 90 days after broadcasting: 16-16-8	Compound fertilizer (N-P-K) - 30 days after broadcasting the seeds: 46-0-0 and 16-20-0 - 65 days after broadcasting: 16-20-0
Gross Revenue (per year)	283,414 THB (1 cycle)	376,696 THB (2 cycle)
Direct Cost (per year)	101,098 THB (1 cycle)	110,367 THB (2 cycle)

Different water regimes also produce different GHG emission profiles for rice cultivation. Irrigated rice produced more emission per meter squared compared to the rain-fed system, due to extended anaerobic conditions under flood irrigation (Wassmann et al., 2000). Neue and Sass (1998) also observed different methane emissions for different rice ecosystems in various countries, including Thailand. Irrigated systems on average produced three times the methane emissions than the rain-fed systems in Thailand. Irrigated rice ecosystems with continuously flooded fields have a longer anaerobic period compared to the rain-fed system, where soil moisture depends on the amount and duration of precipitation.

2.5. Climate Change in the Thai Rice Sector

The agriculture sector is extremely vulnerable to the impacts of climate change. Thailand faced its worst drought in 2010, yet in 2011 more than one million people were affected by severe flooding covering many provinces. Droughts in Thailand happened almost annually in the past decades, leading to critically low levels in water reservoirs countrywide. The extreme changes from drought to flood could happen in the same year, which significantly affects production.

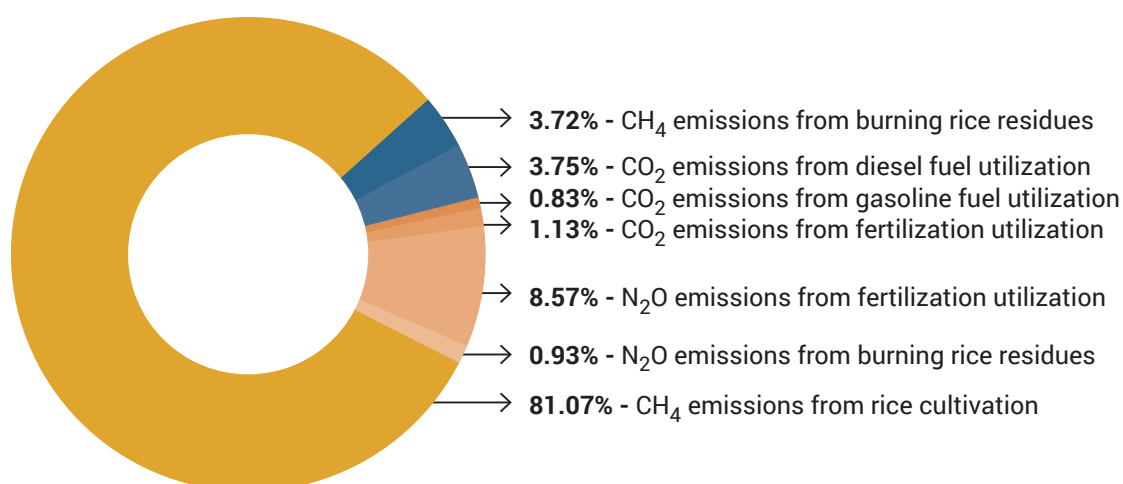
Various government programs supported Thai farmers during the drought period. In the rice sector, the government provides a subsidy to encourage farmers not to plant rice during the dry season to reserve the water (ONEP, 2020).

Rice is highly dependent on water availability which makes it very vulnerable to climate change events. At the same time, rice is categorized as one of the biggest GHG emission contributor in Agriculture sector. Rice is a global environmental concern due to its potent GHG emissions; methane (CH_4), nitrous oxide (N_2O), as well as other environmental issues from land use and application of inputs. The relative impact of GHGs is described by the Net Global Warming Potential (GWP) calculation, which compares the global warming impacts of different gasses in the atmosphere relative to a unit mass of CO_2 . To assess the combined GWP, the magnitude of CH_4 and N_2O emissions are multiplied by their GWP coefficient over a 100-year time. These coefficients are 28 for CH_4 and 265 for N_2O according to the IPCC's 5th Assessment Report. The net GWP for rice cultivation is the summed GWP calculation minus the soil organic carbon sequestered in the rice field (IPCC, 2013; Liu et al., 2013; Zhang et al., 2013). This calculation does not account for the entire lifecycle of rice, such as the embodied emissions of inputs and the energy used for harvesting, drying, storing, processing, transportation, and cooking (Yodkhum & Sampattagul 2016).

Rice fields produce methane during flooded periods, which creates anaerobic conditions that bacteria capable of producing CH_4 to thrive (Wassmann et al., 2000). Rice production also generates N_2O from nitrogen-based fertilizers. Meanwhile, the main sources of CO_2 emissions in agriculture are other cropping operations which release soil carbon or produce CO_2 through fossil fuel combustion such as: tillage, sowing, or harvesting, pumping water, spraying and grain drying. Rice residues burning also releases CO_2 (though this is not counted in the IPCC guidelines, because the carbon was already sequestered from the atmosphere through plant growth) and gaseous pollutants such as carbon monoxide, CH_4 , N_2O , and hydrocarbons in the troposphere (Crutzen & Andreae, 1990).

One study in the Northeast of Thailand assessed the relative GHG impact of different rice cultivation activities (Arunrat & Pumijumnong 2017). This study found that 81% of GHG emissions were CH_4 emissions from the rice production (crop cycle), followed by N_2O emissions from fertilizer utilization, CO_2 emissions from diesel fuel utilization, CH_4 emissions from burning rice residues, CO_2 emissions from insecticide and herbicide utilization, N_2O emissions from burning rice residues, and finally CO_2 emissions from gasoline fuel utilization (Figure 6).

Figure 6. The contribution of GHG emission sources in rice production in the field.



Source: Arunrat and Pumijumnong 2017

Water management in rice fields can significantly reduce the methane emissions. The International Rice Research Institute developed system called Alternate Wetting and Drying (AWD) that can reduce CH₄ emissions by up to 70% compared to a continuously flooded system (Adhya et al., 2000; NAMA Facility, 2021; Prangbang et al., 2020). Management of rice residues can also reduce the GHG emissions, such as returning the rice residues (stubble, straw) back to the field after harvesting rather than burning (Arunrat & Pumijumnong, 2017). This practice, however, requires additional machinery and labour, and in some cases can increase the GHG emissions by increasing the availability of soil organic material under flooding.

Open burning of rice straw is common in the paddy field globally, including in Thailand. Burning releases aerosols such as black carbon, which contributes directly and indirectly to the Earth's radiative balance change, and consequently has an impact on global warming. Black carbon also contributes to the level of Particulate Matter 2.5 (PM_{2.5}), which is a huge issue in Thailand. The Thailand Pollution Control Department (PCD) has monitored the PM_{2.5} level in Bangkok since 2010 (Oanh, 2017). As of December 2020, PCD reported that 34 areas in Bangkok and surrounding provinces have exceeded 50 µg/m³ (micrograms per cubic meter) of PM_{2.5}, which is considered unhealthy. Under the U.S. Air Quality Index, the PM_{2.5} must be lower than 50 µg/m³ to be considered safe. Bangkok was ranked as the 23rd most polluted cities by IQ Air. In this condition, N95 masks are recommended (Thai PBS, 2020).

Most open burning occurs in the central plain of Thailand, in the irrigated rice system. Around 70% of the irrigated rice field area of Thailand practices open burning, whereas only 18% of rain-fed area practices open burning. This is because irrigated land can produce two to three crops a year, so the farmers immediately remove the paddy residues by burning it on the field to prepare for the next crop cycle. Considering that most irrigated rice areas in Thailand still practice open burning; it has affected the increase of PM_{2.5} levels in Bangkok during the rice field preparation period. Kanokkanjana, Cheewaphongphan, and Garivait (2011) estimated that in-field open burning in Thailand at 11,819 ton/year results in around 34,278 tonnes of CO₂ eq. annually. The Thai government has taken various actions to improve air quality, which is further explain in the 20-Year Master Plan on Air Quality Management (2018 – 2037) (ONEP, 2020).

External factors like climate change and natural disasters affect farmers' production in terms of quality and quantity. Consequently, it also impacts cost of production, farmers' income and well-being since farmers must deal with uncertainty in an uncontrollable situation. On the other hand, there are also internal factors which hinders farmers to produce in optimal capacity, such as: farmers' limited knowledge of the latest agricultural technology, financial and bookkeeping management, and mismatch of crop and cropland (Pochanasomboon, Attavanich, & Kidsom, 2020). However, the internal factors can be addressed with technical support and better access to financial instruments. Rice farmers in Thailand have also made efforts towards climate adaptation. The common adaptation practices to climate change found in rice farming were selection of new rice varieties for higher yield and improving soil fertility, while many farmers shifted cultivating dates to avoid the drought hazard in that year (Kawasaki & Herath, 2011).

CHAPTER

3.

SOCIO ECONOMIC
ASSESSMENT

Economically, the agriculture sector accounts for over 30% of total employment in Thailand. However, the contribution of agriculture to Thailand's GDP remained relatively constant at around 8% between 2016 and 2018 (ONEP, 2020). Agriculture includes annual crops, livestock, agricultural services, forestry, fisheries, and aquaculture of which annual crops, especially rice, dominate in terms of land use. The Ministry of Agriculture and Cooperatives (MoAC) has developed several plans and strategies, including the Agriculture Development Plan (2017–2021), National Organic Agriculture Development Strategy (2017–2021), and the Agriculture and Cooperative 20 years strategy. One of MoAC's goals is to boost per capita income for farmer to reach USD 13,000 (THB 416,000) per year.

Thailand's main agricultural products include rice, rubber, and sugar. Globally, in 2019 Thailand was the second biggest rice exporter after India (OEC, "Rice", 2021). Rice accounts for 1.75% of Thailand's total export value, about 4.35 US dollars export revenue in 2019 (OEC, "Thailand", 2021). Around 4.2 million farm households relied on rice cultivation in 2019 (OAE, 2019). The rice sector has significant importance to the socio-economic life of Thai's farming community. Rice is also important to Thailand's and global food security.

This chapter will provide assessment of the socio-economic aspects of rice cultivation in Thailand and of financial instruments and schemes that currently exist to support farmers. Moreover, this chapter will also explore various potential financial instruments that can support Thailand's climate mitigation plan for rice cultivation.

3.1. Rice Farmer Demographics

Agriculture employed 31% of Thailand's labour force with 34% of men and 28% of women in Thailand working in the agriculture sector (ILO, 2019). According to the latest census in 2013, most of the agricultural land holders in Thailand were 35 years old and above, but the holders aged 35 to 44 years old continued to decrease from 26.8% in 2003 to 18.4% in 2013; and the proportion of holders aged 65 years and above increased from 13.1% to 18.5% (National Statistic Office, 2013). The next national census will be carried out in 2023 and it is yet to be observed whether this trend continues.

In the rice sector, the average age of Thai rice farmers was 50 years old, where the average of farming household heads was 61 years (Pattaya Mail, August 15, 2012; Rigg, et al., 2019). The lack of younger rice farmers in Thailand contributes to the declining number of farmers. The younger generation in the farm households find rice farming unattractive. On the other hand, old farmers face declining health and farm work capacity. For example, a survey conducted in Prachin Buri province revealed that some old farmers decided to stop farming and transfer their farm to the successor, some intended to employ additional workers, and some intended to stop farming and dispose of the farmland outside their family or sell it to property developers (Jansuwan & Zander, 2021). This situation puts food security at risk since it may reduce the productivity among farmers and in the region.

According to the last agricultural census, the average farm size for rice farmers was 2.6

hectares per household (Napasintuwong, 2019). The majority of farmers had land titles, and the Central region had the largest proportion of land rentals at 33 percent. Most rice farmers in Thailand grow rice in the wet season, with a smaller share also growing in the dry season. Farmers who have no access to irrigation suffer more financially than other areas (Straits Times, July 28, 2015).

3.2. Farm Economics and Technologies

Declining interest in rice farming is partly due to the low income earned by the farming households. The Office of Agricultural Economics recorded that Thai farmers' annual per capita income averaged 74,483 baht in 2018, about 2,310 USD (Thailand Business News, June 20, 2019). On the other hand, Bank of Thailand's Puey Ungphakorn Institute for Economic Research found in their survey that in 2018 around 40% of farming households were living below the poverty line, set at 32,000 baht a year (Bangkok Post, May 31, 2018). The farm income gap is caused by various factors such as type of crops, farm location, crop cycle, and other factors.

In the case of Ubon Ratchathani, a rice producing province located in the northeast, the average annual income of rice farmers from rice farming is 68,702 baht, and they earn 61,630 baht from other work (Khunthongjan, 2016). Since farmers here only harvest rice once a year, they must have an alternative livelihood to survive in the dry season. In this province, farmers' production-related expenditures were as follows based on a survey conducted in 2013-14: 28% for fertilizer and chemicals (10,325 baht), 27% for labor employment (10,095 baht), 26% for miscellaneous needs, and 19% for agriculture tools (6,872 baht) (Khunthongjan, 2016). To meet the financial needs for purchasing fertilizer and chemicals, farmers borrow money from the bank or informal lenders.

Due in part to the aging rural population, investment in new agricultural machinery has declined since 2000 (Napasintuwong, 2019). This has major implications for adoption of rice mitigation practices, as younger farmers tend to be more eager to adopt and experiment with new technologies. However, adoption of some machinery is increasing due to the declining labor force. Farmers rent equipment such as combine harvesters from agricultural service providers (Napasintuwong, 2019).

3.3. The Thai Rice Market

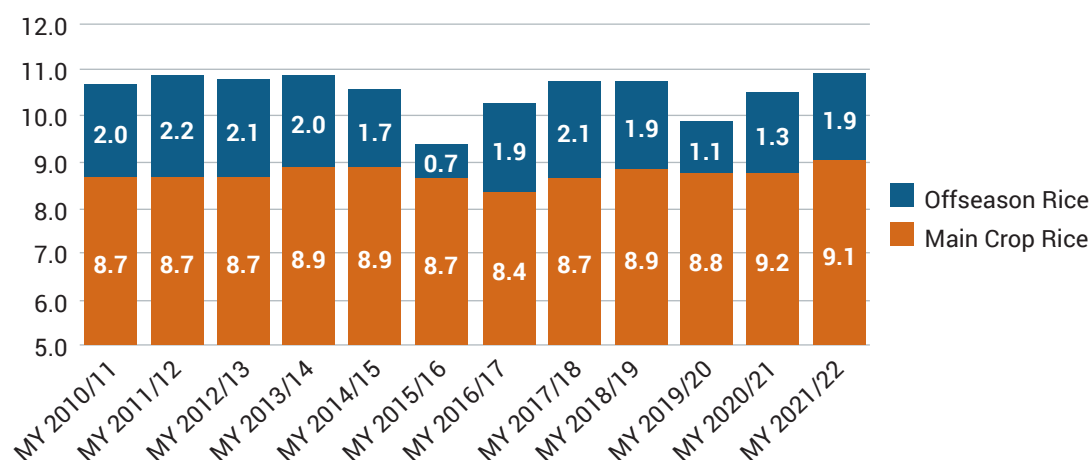
The Thai rice market generates the supply and demand which farmers respond to and influences their income from rice production. About 50% of rice produced in Thailand is consumed domestically and the remaining exported. Exports have remained steady over recent years, but environmental conditions such as flood and drought have impacted the export market because prices rise due to price support systems (Napasintuwong, 2019). Hom Mali rice from Thailand has a good reputation and is in high demand in Asia and in the United States, although recently Cambodia and Vietnam have started exporting fragrant rice varieties. The next sections examine rice production, marketing, and value chain in Thailand more closely.

a. Rice production and trade

The USDA Foreign Agricultural Service predicts that the production in 2021/22 will be 21 million metric tons, and 11% increase from the previous season. Droughts have impacted rice production the previous two years, thus the Royal Irrigation Department encouraged planning of alternative crops such as maize and legumes in water limited areas (USDA FAS, September 21, 2021). Due to increase in production, farmgate prices for paddy rice fell during the first half of 2021. Fragrant and glutinous paddy rice decreased in price significantly while white paddy rice decreased only slightly. Increased acreage will likely continue to depress rice prices (Figure 7).

Figure 7. Thailand's Annual Rice Acreage and Production (M Ha)

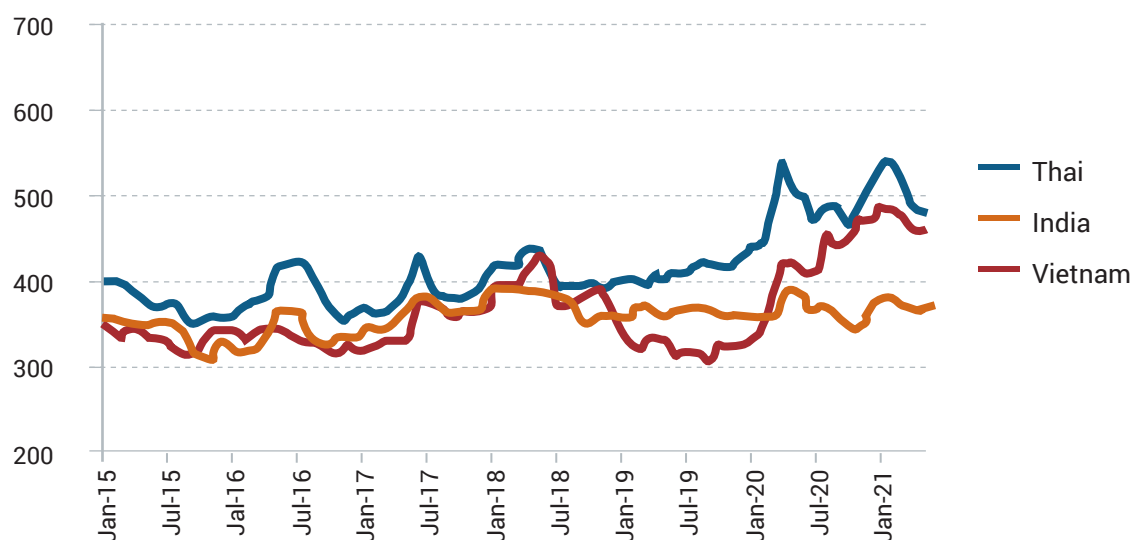
Million Hectares



Thai rice exports are projected to decrease in 2021 compared to 2020, due to shortage of shipping containers and higher transportation costs (USDA FAS, September 21, 2021). COVID-19 has also impacted demand from countries in the ASEAN region. However due to the decreasing strength of the Thai baht against the US dollar, Thai rice prices will lower to become more competitive against Vietnam and India (Figure 8).

Figure 8. Thai, India, and Vietnam rice prices, 25% broken.

USD/tonne



Source: FAO

Image obtained from: https://www.krungsri.com/getmedia/e51e05aa-ae62-4481-9865-6050267d2332/IL_Rice_EN.pdf.aspx

Thai's Ministry of Commerce recorded that in 2020, the United States was the biggest importer of Thai rice, followed by South Africa and then China (Table 2).

Table 2. Thailand Rice Export Destination Jan – Dec 2020 (in kg)

No.	Export Destination	Jan. - Dec.		
		Quantity	Value (THB)	% Share
	WORLD	5,724,659,980	115,914,514,235	100
1	U.S.A.	672,183,490	21,892,387,772	18.89
2	SOUTH AFRICA	672,777,430	9,925,156,642	8.56
3	CHINA	381,363,360	8,426,526,448	7.27
4	BENIN	476,290,010	6,900,826,813	5.95
5	HONG KONG	175,586,272	5,932,466,054	5.12
6	ANGOLA	347,292,330	5,180,910,946	4.47
7	CAMEROON	273,922,147	3,928,872,384	3.39
8	SINGAPORE	127,295,641	3,923,033,411	3.38
9	CANADA	117,460,583	3,653,950,569	3.15
10	JAPAN	257,677,129	3,608,170,721	3.11

Source: Thai Ministry of Commerce

a. Marketing channels

Historically, informal local collectors and middlemen were the traditional marketers in the Thai rice sector. In the past, farmers mainly relied on their services to help find buyers and to help connect them with millers. However, the government began introducing policies which regulate marketing channels, thus the role of middlemen has gradually shifted. The government and the central market of Thailand fully controlled the rice market in the late 1990s. The role then shifted when the Rice Price Guarantee Program (RPS) was revised in 2011, rice millers and agricultural cooperatives collected rice on behalf of the government. Since then, new marketing channels emerged in the supply chain, balancing the traditional channel with a more modernized channel. Although the domestic market in Thailand has been more mature and modern compared to 20 years ago, it is still a regulated market. Any change in the government policy would influence the market and the stakeholders in the supply chain.

In Thailand, there are six main marketing channels which fall into two broad categories: direct and indirect channels (Table 3).

Table 3. Marketing Channels of the Rice Sector in Thailand

Direct Channel	Indirect Channel
Individual Direct Selling From farmers directly to the end-users / consumers	- Rice Miller - Local Collector - Central Paddy Market - Agricultural Cooperative
Group Direct Selling A group of farmers pooled their un-milled rice to be processed and sold to the end-users collectively	

Source: Thamthanakoon, 2019.

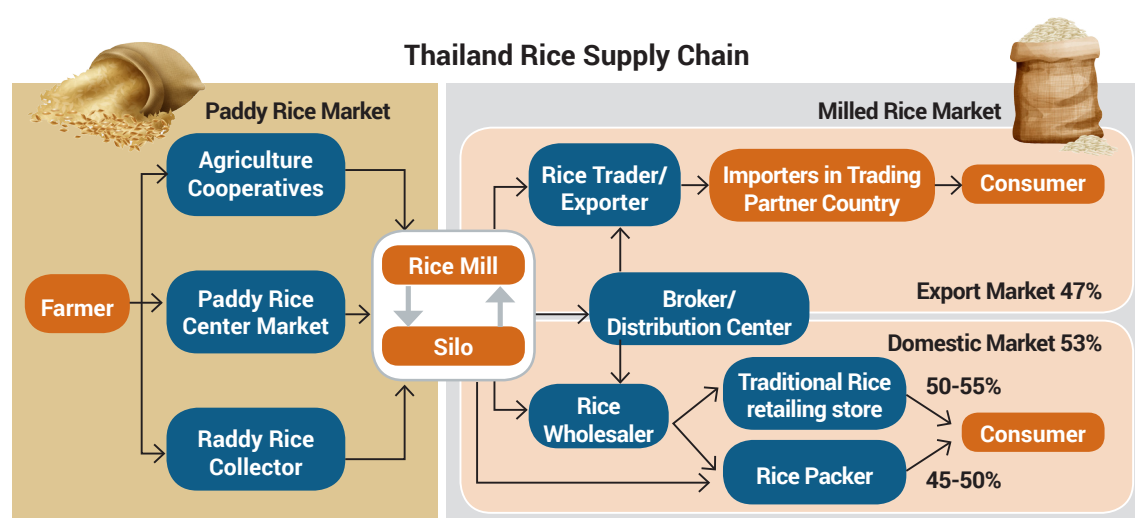
The first category is direct channels which include individual direct selling (in small quantities directly to consumers), or group direct selling where farmers form a group and pooled their un-milled rice to then be processed together and sold to the consumers. The other category is indirect channels including rice miller, local collector, central paddy market, and agricultural co-

operative. Currently rice millers are the biggest intermediary in the value chain in Thailand, they are well-networked among one another, and they provide a shared transportation system which is cost efficient for farmers. Cooperative is also an attractive marketing channel because it offers cash in hand, collective transporting system, and additionally it provides membership benefits.

b. Rice value chain

Farmers are the upstream producers of paddy rice, but downstream actors such as rice millers and trading companies are an important part of the rice value chain in Thailand. Rice mills take unprocessed rice and refine it to milled rice. Thailand has significant rice milling capacity that is higher than the demand, and larger millers have invested in technological improvements that improve the efficiency of the process and quality of the rice (Sowcharoensuk, August 14, 2019). Trading companies then buy rice from the millers or may work through brokers, then rice is packaged and sold domestically or for export.

Figure 9. Rice supply chain in Thailand.



Source: Sowcharoensuk, August 14, 2019.

Table 4. Thailand's key rice value chain players (as of 2018).

Rice Mills		Packed Rice		Exporters		
		Company	Brand	White Rice	Jasmine Rice	Parbolled Rice
• Ake Rice Mill	• Ponglarp	• Asia Inter Rice	• Benjarong and Suphannahong	• Ake Rice Mill	• Asia Golden Rice	• Asia Golden Rice
• Asia Golden Rice	• Riceland International	• Chia Meng Marketing	• Hong Thong	• Asia Golden Rice	• Bangsue Chia Meng Rice Mill	• Capital Rice
• Capital Cereals	• Sangfah Agri Product	• Chaithip	• Phanom Rung	• Capital Cereals	• Capital Rice	• CP Intertrade
• Capital Rice	• Thai Capital Crops	• Chaimongkol Rungrueng Agriculture	• Bua Thip	• Capital Rice	• CP Intertrade	• Golden Granary
• Chia Meng Marketing	• Thai Fah (2511)	• CP Intertrade	• Royal Umbrella	• Ponglarp	• Herba Bangkok	• Premium Rice Export
• CP Food Store	• Thai Grandlux International Rice	• Patum Rice Mill and Granary	• Mah Roonkrong	• Sangfah Agri Product	• Siam Grains	• Riceland International
• CP intertrade	• Tong Hua Bua Yai (1994)	• Sandee Rice	• Sandee	• Thai Capital Crops	• Thanasan Rice	• Sangfah Agri Product
	• Vudhichai Produce	• Singha Corporation	• Pun Dee	• Thai Grandlux International Rice	• Universal Rice	• Sawatpaiboon Rice Mill
		• Soonthornthanyasup	• Kaijaa	• Thanasan Rice	• Vudhichai Produce	• Thai Granlux International Rice
		• Thai Ha	• Kaset			

Source: Sowcharoensuk, August 14, 2019

3.4. Access to Finance

Access to finance is key to the development of the agricultural sector including rice cultivation. Over the last few decades, the Thai government has pushed for increased agricultural finance, resulting in now around 90% of farming households having access to agricultural credit (Napa-sintuwong, 2019). However, since there are factors that pose risks to lenders such as climate and disaster risks, coupled with the fact that agricultural commodities are highly sensitive to price volatility, it makes formal lenders like banks cautious when giving credit to farmers. Farmers must meet certain qualifications to be eligible to receive credit from banks. One of the key factors which determines farmers' bankability is land ownership. Pochanasomboon, Attavanich, and Kidsom (2020) categorize land ownerships into two categories: full land ownership and weak land ownership. Full land ownership means the farmer has the exclusive right to own and utilize the rent, the right to transfer or to sell the property to somebody else and has the security over encroachment (enforceability rights). On the other hand, weak land ownership means the farmer only has the exclusivity and enforceability rights but not the right to sell nor to transfer it to other people. There are different types of certificates of land tenure in Thailand, some present the right of farm households to utilize the land, but the ownership of the land is not attached to the farm households, rather the ownership status belongs to the government.

In Thailand 59.3% of agricultural households fully own their lands, which is consistent with the well-defined property right structure in 2017 (Attavanich et al., 2020). Research revealed that land with full ownership has a higher value in the market, making the farmers eligible to have a higher bank loan (at least 80% of land value) compared to the farmers with weak ownership (at most 50% of land value) (Pochanasomboon, Attavanich, & Kidsom 2020). Thus, farmers with weak land ownership, especially for those with small and medium size land, have more limited access to agricultural credit.

Moreover, some smallholder farmers do not own the land at all but rent the lands from the owners. According to the Office of Agricultural Economics (2017), around 19% of agricultural land is rented and this rate has been stable over the past decade. Land rent schemes put farmers in a disadvantaged situation and without having legal ownership of the land. Tenant farmers are more reluctant to invest in technology, appliances, and/or new systems to improve their farms since rent contracts are usually short term. Another traditional scheme which still exists is to rent land from families without any written agreement, for example this practice can be found in Prachinburi province (Faysse, Phiboon, Aguilhon, Srisawaluck, 2020).

Thai Bank for Agriculture and Agricultural Cooperatives (BAAC) has played an important role in providing financial services to small holder farmers in Thailand since 1966. The establishment of BAAC has diminished the role of informal lenders in the villages who lent money at high interest rates to farmers. Currently, BAAC offers several types of loans to individual farmers. There are different types of financial services and supports provided by BAAC for farmers in Thailand, including: loan, pledging scheme, price guarantee, and crop insurances.

a. Loans

BAAC has seven types of loans for different purposes and different terms of payment, dedicated to support farmers to meet their financial needs (Table 5).

Table 5. Type of Loans from BAAC

Type of loan	Description	Term of payment
Short Term Loans for Agricultural Production	The objective of this type of loan is to meet production costs during a given production season, such as costs of land preparation, seeds, fertilizers and labor hiring.	12 months. In extraordinary cases, it can be 18 months.
Loans for the Postponement of the Sale of Farm Produce	The objective of this type of loan is to help farmers meet their household expenses while they postpone the sale of produce to avoid selling during periods of over-supply and low prices.	6 months.
Medium Term Loans	Medium term loans are intended for investments in agricultural assets that can be used for more than one year, for example loans for investment in agricultural land reclamation or improvement or the purchase of agricultural machinery or livestock.	3 years, in extraordinary cases it may be extended to 5 years.
Cash Credit Loans	A cash credit line is a kind of short-term loan for crop production, which provides great convenience for farmer clients.	Once a farmer has entered into a cash credit agreement, he or she can make several withdrawals up to the loan ceiling and within the period of the agreement which will not exceed 5 years.
Long Term Loans for Refinancing Old Debts	This type of loan is for the redemption and repurchase of agricultural land that originally belonged to the farmers, their married partners, children, or parents. These loans are intended to maintain the farmers' land ownership. They are also provided for agricultural expenses of the first planting season, investment in necessary farm assets and expenses involved with mortgage of immovable property.	
Long Term Loans for Agricultural Investment	This type of loan is intended for investments in fixed agricultural assets to improve existing production enterprises or introduce new ones. These investments are relatively expensive and require a long period of time before the enterprises can be expected to break even. This type of loan can be available to individual borrowers and to farmers in special projects.	Within 15-20 years. A grace period for the repayment of principal and/or the payment of interest may be allowed but normally not for more than the first five years.
Loans for Farm-Related Activities	This type of loan is intended to meet expenses and/or investment costs for activities related to agriculture. These activities use agricultural produce, either from farmers themselves or provided from other sources, as raw materials for processing as finished or semi-finished goods for sale. Qualifying activities may be for either production or services related to agriculture. This type of loan may be classified into two types: short term and long term loans.	Short term loans for production which are intended to meet the production costs of farm-related activities. Repayment of these loans is normally required within twelve months. Long term loans for investment in assets used in farm-related activities. Repayment of these loans is normally required within fifteen years, or in twenty years in extraordinary cases.

Source: BAAC Credit Services

b. Rice Price Guarantee Program

In addition to these loans, BAAC also disburses direct government subsidies for rice farmers. Thailand's rice sector has long been a heavily subsidized sector. Between 2011 and 2014, the government implemented a rice pledging scheme. This scheme had two elements, firstly, the government made a direct purchase of rice with a price higher than the prevailing market price. Secondly, exports were reduced to influence the international market price (Warr, March 17, 2014).

The government reformulated the price support scheme and introduced a rice price guarantee programme in 2019. The scheme was applied when the market price dropped below guaranteed price, the price difference would be covered by BAAC. The government set different prices, conditions, and quota (volume) for different types of rice (FAO, 2019).

The rice price guarantee programme continued in the 2021/2022 season and the government also provides a loan for farmers to delay selling their paddy. This measure is relevant for a specific type of rice, e.g., Hom Mali rice, whose peak of harvest is between November and February. By delaying the sales and slowing down the supply in the market, this program helps to avoid sharp price fall during the harvesting months.

c. Crop Insurance

Crop insurance is one of the financial instruments that the BAAC offers to farmers to protect their farm particularly from natural disasters such as flood, drought, and/or disease. Since the country has been suffering from catastrophic climate disasters in the past decade, the government issued crop insurance schemes for key commodities that are particularly vulnerable to climate conditions such as rice. The government began offering financial support for farmers that were affected by natural disasters through a scheme called Government Disaster Relief in 2005, operated by the Ministry of Agriculture and Cooperative (MoAC). The support was THB 1,113 per rai⁵ for rice crop, and it was fully funded by the government. In 2011 the government introduced a crop insurance scheme which was only partially subsidized, and the farmers had to start paying premiums. The Government Disaster Relief was discontinued in 2013 due to insufficient funds and the crop insurance continued, mainly covering drought, flood, storm, or typhoon, cold, hail and fire. In the following year the scheme also began to offer insurance protection against pests and diseases.

Since it was first established, the crop insurance scheme operation in Thailand has evolved and undergone several administrative changes, being from a 100% public operated insurance scheme into a public-private insurance scheme. Currently, the crop insurance system is managed by Fiscal Policy Office, District of Agricultural Extension, BAAC, Office of Insurance Commission, Thai General Insurance Association, Department of Disaster Prevention and Mitigation), 24 insurers and 13 reinsurers companies. The government formulated the policy and premium rating criteria, and insurers (including private) are obligated to offer insurance to all farmers in the region to qualify for the premium subsidy. Subsidy range is up to 100% (depending on the level of risks).

There are two types of insurance payment systems that exist in Thailand (Sinha & Tripathi, 2014). The first type is the index-based payment, which is verifiable, including independent measurement of variables that impact crop development. This has been applied through a weather-based index system. Secondly, indemnity payment which means the pay-out is based

⁵Rai is a commonly used measure of area in Thailand, equivalent to 1,600 square meters or 0.16 hectares.

on measured and/or declared losses. The current system that is in place is a zoning classification system (low-moderate-high risk), based on risk assessment from historical data, which works the same way with index-based payment (payment rate already fixed in advance).

The implementation of crop insurance in Thailand has also been combined with other initiatives. For example, combining crop insurance with technological development that helps farmers to monitor their farm growth, assess and manage the risks. In 2015 BAAC cooperated with GIZ through the RIICE (Remote Sensing-based Information and Insurance for Crops in Emerging Economies) project with the aim to reduce the vulnerability of smallholder rice farmers using remote sensing technologies that could map and observe rice growth.

3.5. Stakeholder Analysis

Based on on-going rice projects in Thailand, various stakeholders that could help the transformation of the rice sector in Thailand to be more sustainable were analysed and identified. The information below is gathered from available project documents, website from each institution, and discussions with related stakeholders.

Table 6. Stakeholder Roles and Functions

Name of organization	Role	CC related functions/ projects
NATIONAL GOVERNMENT AGENCIES		
Ministry of Natural Resources and Environment (MoNRE)		
Office of Natural Resources and Environmental Policy and Planning (ONEP)	Main implementing partner responsible for climate change mitigation topics in Thailand. ONEP is also responsible for monitoring and administration of NDC implementation in Thailand, and other ministries (including MOAC) report their sectoral implementation/ progress to ONEP	On climate change: Forest and biodiversity management, water management, solid waste and environmental management, eco-friendly production and consumption, reduction of CC and natural disaster effect
Pollution Control Department (PCD)	Responsible for policy development, planning, standards setting, intra-agency and international cooperation, audit, and evaluation of pollution-related issues	PCD is Thailand's focal point for the Climate and Clean Air Coalition. PCD's workstream include waste management, hazardous substances, water quality, air, noise and vibration levels.
Thailand Greenhouse Gas Management Organisation (TGO)	Autonomous government organization to support GHG reduction projects such as emissions trading/carbon market.	Manages Thailand's voluntary emission reduction platform, T-VER, and Thailand's carbon credit registry. Supports capacity building for GHG inventory and MRV activities.
Ministry of Agriculture and Cooperatives (MOAC)	Responsible for all agriculture-related matters, including water sourcing and irrigation development, agriculturist promotion and development, cooperative system promotion and development, including production process and agricultural commodities	

Thai Rice Department (TRD)	Main agency responsible for policy, strategy, and international cooperation of the rice sector in Thailand. Lead organization in rice production R&D, value creation, and development of sustainable rice production. There are some programmes that are initiated and maintained by TRD, which help later reach-out projects to farmers: • Mega Farm Projects • Rice Seed Bank Project • Organic Rice Project • Rice Research and Development Project • Integrated Organic and Rice Market Linkages Project • Smart Farmer Project • Rice seed production project • Community Rice Seed Center project • Niche market rice project	Key provider of technical support on sustainable rice production (organic, Good Agriculture Practices (GAP), GAP++, SRP) as well as GHG emissions mitigation technology, and cooperates with relevant government and private agencies at the local level in providing extension services to farmers groups and communities. RD is also in charge of SRP at the national level, also host of SRP's Project Management Unit (PMU) Develop and coordinate intra-agency policy that support the integration of SRP; support forest, land, and water management enhancement; encourage recognition, adoption and scale up of SRP.
Office of Agriculture Economics (OAE)	Lead organization in developing and formulating country's agricultural strategy as management tools for security, prosperity and sustainability, responsible for official agricultural data set, National Agriculture Big data Centre	As the secretariat of the Committee on Climate Change under the MOAC in drafting the Agricultural Climate Change Action plan that aligns with the Agricultural Strategic Plan on Climate Change (ASPPC) for the period of 2017-2021. OAE is also in charge of overseeing projects implemented by the ministry.
Royal Irrigation Department (RID)	Responsible for water management related to agriculture, energy, household consumption, or industrial purposes	Provide advice and manage irrigation water management activities, Alternate Wetting and Drying (AWD), and watershed management.
Department of Agricultural Extension (DoAE)	Transferring agricultural technology and extension services to farmers, agricultural data collection via Farmbook and Registration Agriculture activities system	At the national level: in charge of the reduction of agrochemical pollution through regulatory approaches. At provincial level: support the expansion of SRP Standard practices land through capacity development
Land Development Department (LDD)	This agency is a main national coordinating unit for UNCCD. Its key duties include land classification, utilisation, establishing conservation areas, and promoting soil fertility, agricultural productivity and long-term sustainability, water management by a project for providing small pond to small farmers	In charge of spatial landscape management plans that are integrated with sustainable agriculture and environmental conservation; and provide technical support for fertilizer and soil management (for farmers)
National Bureau of Agricultural Commodity and Food Standards (ACFS)	Standard setting for agricultural system, commodity, and food safety; accreditation of certification bodies, trade-related issues such as SPS/TBT, and standard compliances	Integrate the SRP Global Standard into the National (Thai) Good Agricultural Practices (GAP) for rice, which also include some climate mitigation actions together with the SRP National Chapter for Thailand.

FINANCIAL INSTITUTIONS		
Bank of Agriculture and Agricultural Cooperatives (BAAC)	Bank that provides financial assistance to farmers, farmers associations, and agricultural cooperatives which conduct agri-related business	Manage various financial instruments aim at climate adaptation (e.g., loan, crop insurance) and mitigation (e.g., green bonds for forestry and environmental conservation projects, which include organic agriculture)
STAKEHOLDER INSTITUTIONS		
National Community Rice Centre (CRC) Network	Organization of cooperative structures (Community Rice Centres) to facilitate outreach to rice farmers in Thailand on an aggregated level	Support policy and program implementation on the ground, including socioeconomic, environmental and livelihood activities; also, key point of contact in development of research throughout Thailand.
Thai Rice Exporters Association	Representing interests of rice exporters in Thailand in cooperation with the project	Enhance business operations aiming ultimately at the principles of good governance and social responsibility, leading to sustainable economic development.
SRP - National Chapter	A national level of SRP multi-stakeholder platform that actively promote the SRP standard implementation in Thailand. The global SRP is co-convened by UNEP and the IRRI and is managed by its Secretariat located in Bangkok.	SRP project implementer and partnership coordination in Thailand. Its mandate includes policy coordination, standard implementation, knowledge management, and monitoring and evaluation of all SRP-related activities in Thailand
RESEARCH INSTITUTIONS		
Joint Graduate School of Energy and Environment (JGSEE)	The Joint Graduate School of Energy and Environment is an autonomous graduate school, operates as a consortium of five Thai universities and located in King Mongkut's University of Technology Thonburi (KMUTT). Established in 1998, aims to be an internationally recognized premier centre in graduate education and research in the fields of energy and environmental technologies	Laboratories in Advanced Greenhouse Gas and Aerosol Research and Energy and Environmental Policy. Researchers at JGSEE/ KMUTT are developing an application for digital MRV of agricultural GHGs, among other work in climate mitigation and agriculture.
Kasetsart University	Kasetsart University has a department specific for rice research called Rice Innovation Centre of Excellence, which focuses on genomics, rice gene discovery, and molecular breeding.	Research and develop climate-ready crops, including rice. The research is aimed at improving the resilience of rice crop in a changing climatic condition

International Rice Research Institute (IRRI)	International research partner to help improve the life of rice farmers and consumers, promote sustainability	Collaborate with Thai Rice Department for climate mitigation in the rice sector through NAMA prototype, led climate-smart agriculture program namely CURE (Consortium for Unfavourable Rice Environments), and support the government in sustainable pest management work. IRRI has also developed the SECTOR tool to calculate the GHG emission in the rice sector and other Information and Communication Technology (ICT) tools such as Farm Management that could be used further in the project.
INTERNATIONAL AGENCIES		
GIZ	GIZ Thailand is well established as a lead organization in food and agriculture development in Southeast Asia. Rice and climate are key topics in GIZ's work in Thailand	Manage and implement Thai German Climate Program (TGCP), Thai Rice NAMA, Market-Oriented Smallholder Value Chains (MSVC), SARI. GIZ has implemented SRP Standard in its rice projects in Thailand since 2014. In addition, GIZ was one of the co-conveners of SRP globally, and the host of the SRP National Chapter – Thailand. GIZ was endorsed to be the Executive Agency (EA) on behalf of TRD for the proposed GEF - Inclusive Sustainable Rice Landscapes project.
FAO (Food and Agriculture Organisation)	FAO supports Thailand mainly in three priority areas, namely: 1) implementing food safety and standards, 2) improving agricultural value chain opportunities and 3) improving sustainable natural resources management	Co-manage 'Integrating Agriculture in National Adaptation Plans' or NAP-Ag project with UNDP. The activities include updating Thai NAP, monitoring and assessment of adaptation actions, capacity building, and improving the overall NAP processes. FAO also part of the GEF Inclusive Sustainable Rice Landscapes project.
UNEP	UNEP coordinates global environmental activities and assist developing countries in implementing environmentally sound policies and practices. Bangkok is UNEP's Asia Pacific headquarter. UNEP's area of work in Thailand includes sustainable agriculture, green economy, biodiversity, and many others	Main organizer for upcoming GEF Inclusive Sustainable Rice Landscapes project in Thailand, and technical expert for TEEB AgriFood program. UNEP was a previous co-convenor of Sustainable Rice Platform and is Implementing Agency of the proposed Sustainable Rice Landscape's GEF project in Thailand

SRP Global Secretariat	The Sustainable Rice Platform (SRP) is a multi-stakeholder alliance with over 100 institutional members from public, private, research, civil society, and the financial sector. SRP's aim is to promote the SRP Standard for Sustainable Rice Cultivation, which includes climate-friendly rice cultivation practices. The Global Secretariat, with its headquarters in Bangkok, promotes adoption of the SRP Standard globally and engages value chain actors in the private and the public sector.	Provide global voluntary standard for sustainable rice production which is complemented by a set of performance indicators and an assurance scheme.
PRIVATE SECTOR ACTORS		
Olam Int.	Olam International is one of the leading global agri-business actors involved in the whole rice value chain in Thailand. Olam works to support smallholder farmers in improving crop quality and yield to provide customers with consistent volumes of sustainable products	Olam International partners with the Thai Rice Department, UN Environment Programme (UNEP), GIZ/NAMA Facility, and IRRI on the Sustainable Rice Platform (SRP) as a governing member from private sector to define internationally accepted, scientific solutions to the climate impacts of rice agriculture.
Herba Thailand (Ebro)	Herba/Ebro is a leading rice group actively involved in the international rice value chain/trading in Thailand (headquartered in Spain). It has its SDG goal to transform their supply chain to be more sustainable, including their rice sourcing business.	Sourced the Sustainable Hom Mali Rice production from 2018-2020 along with Mars Food and GIZ in Roi Et province (SARI project).
LOCAL NGO (Non-Governmental Organisation)		
Green Net	A social business organisation which promotes organic agriculture and fair trade in Thailand and beyond	Develop and scale up South-South Climate Adaptation program, Community Based Adaptation for Food Security in Thailand, Rice Chain Climate Change Adaptation Program (in Yasothorn), Organic Agriculture Production and Certification, and many others

CHAPTER

4.

INTERVENTIONS TO IMPROVE
THE THAI RICE SECTOR

A range of ongoing initiatives aim to improve the Thai rice sector. These range from economic support of farmers, to development of value chains, to promoting adoption of low-emission practices. This chapter outlines some of these existing interventions in the Thai rice sector.

4.1. International Cooperation Projects in Thailand

The Thai government works together with international partners, tapping into the international resources and sustainable agriculture expertise. One of the flagship cooperation projects is the Thai Rice NAMA (Nationally Appropriate Mitigation Action), which is funded by the NAMA Facility. Another noteworthy cooperation is Thai-German Climate Programme – Agriculture (TGCP-Agriculture) which will be concluded in December 2021. The Market Oriented Small-holder Value Chains (MSVC) is another agricultural project, which promotes the SRP standard practices to strengthen the economic sustainability of the farmers by strengthening rice value chains. Moreover, there are several projects that are prepared by various international agencies in Thailand. One of it is the GEF Inclusive Sustainable Rice Landscapes project that is led by UNEP. The various projects are listed in Annex 2, and further information on these projects are described below.

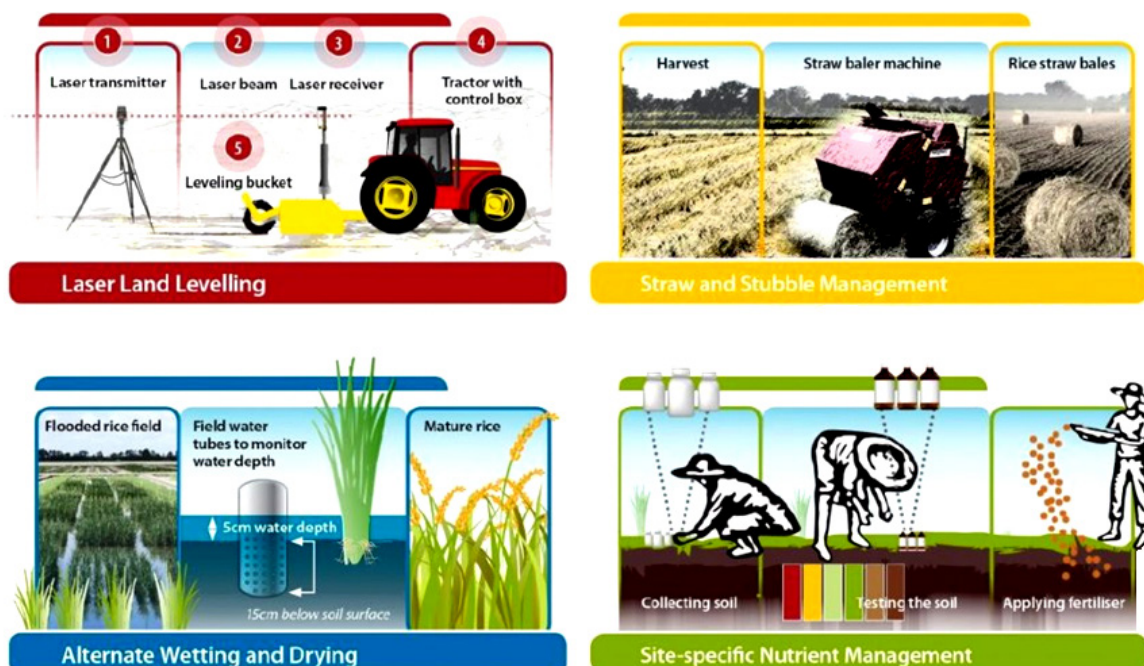
1. Thai Rice NAMA

Launched in 2018, this project supports 100,000 irrigated rice farmers in Central Thailand to implement low-emission rice farming and provides them with access to mitigation services and technologies. The program is funded by the NAMA Facility, an initiative of Germany, UK, Denmark, and European Commission. The Thai Rice NAMA aims to reduce GHG emission from rice farming. There are two main mitigation technologies that are promoted in this project, which focuses on water management. The Thai Rice NAMA promotes adoption of AWD in combination with the Laser Land Levelling (LLL) to maximize the water management system (Figure 11). The program also promotes financial incentive mechanisms to facilitate a sector transformation to mitigate rice sector emissions by scaling adoption of the SRP Sustainable Rice Cultivation Standard (SRP Standard). The project is still ongoing and is expected to conclude in 2023.

The project provides three intervention strategies to help Thai farmers shift towards low-emission rice production:

1. **Low-Emission Rice Production Technology:**
Providing effective mitigation technologies and establishing a Revolving Fund to support the start-up costs for mitigation service provision.
2. **Mitigation Technology Services:**
Leveraging national green credit program for service providers to provide mitigation technology that requires huge investment, such as laser levelling, and straw or stubble management.
3. **Policy Formulating & Supporting Measures:**
Supporting the government in scaling up and integrating the mitigation solutions into the Thai government's workplan, projects and budget at all levels

Figure 11. Four Basic Mitigation Technologies of Thai Rice NAMA



Source: ASEAN – Agrifood 2018

2. Thai German Climate Program (TGCP) – Agriculture sector

The TGCP programme supports the Office of Natural Resources and Environmental Policy and Planning (ONEP) to implement the Thai NDC by improving the framework condition. Five sectors are targeted by this project: national climate policy, renewable energy, waste management, agriculture, and water (irrigation). One focus of this project is to improve the MRV (Measurement, Reporting, and Verification) system for GHG emissions and mitigation,⁶ and to improve the Monitoring and Evaluation (M&E) System to assess adaptation interventions. The program should ensure that all approaches are compliant with the Thai authority and aligned with the UNFCCC reporting requirements (GIZ, 2019a).

In the Agriculture sector, the programme partners with the Thai Rice Department to introduce interventions which aim to mitigate the rice sector's GHG emission. The areas of interventions include: developing national MRV system, conducting nationwide train-the-trainers program on SRP Standard implementation, developing Thai GAP++ standard for rice which aligned with the international SRP Standard, developing institutional structures for mobilizing funds for NDC implementation in the rice sector, and supporting policymakers in formulating a potential GHG emission reduction target for the agriculture sector through workshops and conferences. The program targeted 15,000 farmers to be trained in sustainable and low-emission rice farming in the northeast and central of Thailand (GIZ, 2019b).

With the responsibility of ONEP for submission of the GHG Inventory Report, it the TGCP also aims to improve the inventory data system for agriculture. The mitigation estimation system for agriculture was developed in collaboration with IRRI, and it uses the SECTOR tool to calculate the GHG emissions from the rice sector. The project has also provided the SRP Standard toolkit in Thai Language as an initial step to introduce the SRP Standard not only to farmers, but also to the officials, as a first step to developing a Thai GAP++ Standard for rice.

⁶See <https://www.asean-agrifood.org/thai-government-measures-local-greenhouse-gas-emissions/>

3. Market Oriented Smallholder Value Chains (MSVC)

The MSVC project is implemented from 2018 until 2022, targeting 16,000 small holder farmers in Thailand with an aim to improve the sustainability smallholder value chains through training and adoption of the SRP standard. The project is co-funded by the BMZ (German Federal Ministry for Economic Cooperation and Development) and Crop Life. Olam International and Crop Life are the implementing partners along with the Thai Rice Department. This project is a continuation of the previous Better Rice Initiative Asia (BRIA) project that was implemented from 2014 to 2018. Both projects were implemented not only in Thailand, but also in Viet Nam, Indonesia, and the Philippines (the MSVC however is not implemented in the Philippines). From 2018 until 2020, the MSVC in Thailand has trained more than 10,000 farmers in Ubon Ratchathani and Surin provinces with the SRP standard for Sustainable Rice Cultivation practices. The project aimed to produce 60,000 tonnes un-milled Thai Hom Mali Rice that comply with the SRP Standard by 2020. Olam as the main buyer has supported the project to distribute the rice and to support the third party standard verification process of 4,290 smallholder farmers in Ubon Ratchathani and Surin which covers around 11,900 ha of rice paddy (GIZ, 2020).

4. Sustainable Aromatic Rice Initiative (SARI)

The Sustainable Aromatic Rice Initiative (SARI) ran from 2018 to 2020. SARI aimed to collaborate with 1,200 Thai farmers in Roi Et, a province in Northeast Thailand, to strengthen their farmer group and provide capacity building to adopt the SRP Standard. The project trained and supervised the farmers to produce rice that complies with the SRP Standard and to facilitate the market linkages of a targeted 3,500 MT Hom Mali rice with local millers and exporters. The project was a collaboration between GIZ with Mars Food and HERBA (Ebro) (Askew, July 18, 2018; GIZ, 2019c). From the discussion with HERBA, this project was a commitment from HERBA and also Mars Food to reach their SDG target, and to increase their sourcing of sustainable products.

5. Global Environment Facility (GEF): Inclusive Sustainable Rice Landscapes

The project is planned to commence in 2021 and conclude in 2025. The project plans to bring a landscape approach to transform rice production in Thailand by integrating competing social, economic, and environmental interests through policy, capacity building, financing, and sustainable land management practices. It is proposed under the GEF Food Systems, Land Use and Restoration (FOLUR) Impact Program. The project is targeted for North Thailand – Chiang Rai province – and Northeast Thailand – Ubon Ratchathani province.

6. TEEB AgriFood Program (The Economics of Ecosystem and Biodiversity)

The TEEB programme has been led by UNEP since 2008 and focuses on research and capacity building on holistic evaluation of eco- and food systems value chains. The programme in Thailand is implemented under the International Climate Initiative (IKI) and under the European Union Partnership Instrument (EUPI). The research is led by a team from Khon Kaen University, Thailand, comparing the organic and conventional rice production practices in Buriram and Surin Provinces – in the Northeast of Thailand. The research focuses on the economic aspects of adoption of various standards, such as organic standard and SRP Standard, and how these standards could promote biodiversity and ecosystem conservation. Additionally, TEEB conducts analysis at the landscape level by using spatial models to generate results at a local or regional scale (e.g., watershed level). In this way, analysis would consider landscape configuration and context as these are key factors in determining impacts on many ecosystem services and biodiversity (TEEB, 2020).

4.2. Rice program by Thai Government

The Thai government has established various projects and allocated substantial budget to the rice sector. According to the Thai Rice Department (TRD), one of the most successful approach from the MoAC is the “Mega Farm programme,” which was designed to synergize various program or activities from different government agencies and private sectors to farmers through organizing farmer groups. The Mega Farm programme aims to enhance the resource mobilisation by stakeholders along the value chain through the farmer groups. A field manager or committee acts as the focal point for managing the farmers’ organization activities along the rice value chain and encourages the members to work together as a group when managing their inputs, cultivation, quality assurance, and market linkages. The scheme contributes to reduced production costs, improved yields, enhanced bargaining power, increased competitive advantage, and enhanced farmer group capacity. A typical Mega Rice Farm consists of 100 to 150 farmers with a planting area around 480 hectares. The Thai Rice Department also has encouraged farmers to work in a group, called Community Rice Center (CRC). CRC was formed to encourage farmers to produce certified seeds. Some CRCs were combined to form Mega Rice Farms.

The TRD also promotes agricultural projects such as the: Conditional Incentive Project, Precision Farming and Market Linkage, Mega Farm Project, and the ICT project. The objective of these projects is to improve productivity, reduce production cost, and to improve rice product quality. The projects’ approaches try to encourage farmers to design their own project planning and to organise the budget wisely. Two of these projects are highlighted below:

1. Mega Farm with market linkages (COVID-19 release)

Unlike the usual mega farm project, this program was established in 2020 with a budget around 70,000 million THB. Certain budget is allocated to the groups and the farmers are given the independence to design their project. Although it has the same set up with the mega farm programme, this COVID-19 release program has a greater budget and allows farmers to design their activities more freely. The program is monitored by MoAC. Most of farmers focused the program on farming machinery for land preparation and seed drying.

2. ICT projects

Agricultural data is very important and valuable to help farmers to improve their practices and increase productivity and crucial for the government to prepare new programs to support farmers. Currently the OAE is in charge to organise the data, however, there are some rice farming data that is organized by the TRD.

It is important that the CCAC project in Thailand align with these existing programs as well as future planned projects in the agriculture and climate sectors.

4.3. Lessons Learned

The researcher discussed the research detailed in this report with the UNEP project coordination team and conducted a desk analysis to understand the challenges faced by the project that could be insightful for the CCAC project. These lessons are listed in the table below.

Table 7. Lessons learnt from various project implementation in Thailand

No	Factor	Lessons Learned
1	Global Pandemic	The global COVID-19 pandemic has affected programme implementation, e.g., limited ability to provide trainings and to monitor the project implementation on the project sites
2	Outreach model and implementation partner	Introducing a new set of technology requires long-term engagement and an adaptive management and learning approach. Even if the technology is not new, there is always a transition process to encourage farmers to adopt a new technology or management approach. Technology promotion projects require strong outreach capacity that engages various stakeholders. Considering the current UNEP capacity in Thailand, it might be better to engage with another agency that already has a strong presence on the ground, such as an Extension Office, local NGO, or other multi-national projects with TRD (e.g., Rice NAMA project)
3	Climate Event - Drought - Flood	There have been extreme climate issues in Thailand due to alternating drought and floods in the past years that affected rice production, making it difficult for the rice trader to plan an effective market plan (Bangkok Post, July 23, 2020; NNT, 2020). From the Rice NAMA project, the project coordinator mentioned that the water availability also affected farmers in adopting the AWD technology. Farmers feel insecure to irrigate their farm fields, knowing that there is no constant water supply due to the drought. Government also gave incentives for farmers in the irrigated area not to plant during the dry season due to the extreme drought in 2019. Crop insurance could give more assurance for farmers to practice AWD despite the climate. Additionally, as TRD is developing a new rice variety, it could help farmers to reduce their water use.
4	Financial Model	Thai Rice NAMA project has brought a new financial model in Thailand through the Revolving Fund and green credit for the farming service provider. Further assessment of this financing model should be conducted. In addition, the upcoming GEF Inclusive Sustainable Rice Landscapes project also assesses several financial model alternatives for Thailand with the Rice Bonds that have been initiated by BAAC (See Chapter 3). Another lesson could also be taken from the RIICE project that offered rice crop insurance.
5	Monitoring System	Two types of MRV data are needed in Thailand: national GHG inventory data and project-level mitigation data. Thailand is committed to collect only the inventory data for the agriculture sector. This should prompt the project strategy to adjust the monitoring system according to the government's preference.
6	Policy Alignment	Finding the right government focal points to support the implementation the project is very crucial. International projects should also be flexible to adjust the project objectives to align with the government strategies and priorities. Having a strong infrastructure to build a good relationship with the government and to have frequent policy dialogues to align the project would be needed.
7	Market Value	Both Olam and HERBA (the buyer of the SRP rice in Thailand) have admitted that despite the good quality of the SRP rice, there is limited awareness from the market on sustainable rice production. Being a staple food that is mostly consumed by the producing countries, the rice value chain has different characteristics than other agricultural commodities such as coffee or palm oil. The upcoming project could try to raise awareness in the Thai and global rice communities to create a market-pulling effect that would motivate farmers to adopt the recommended practices. Moreover, the Thai Rice Department also mentioned that the SRP rice should be promoted as GAP ++ rice, considering the familiarity of GAP rice in the market and the Thai rice communities. It is not advisable to introduce a new term into the market.

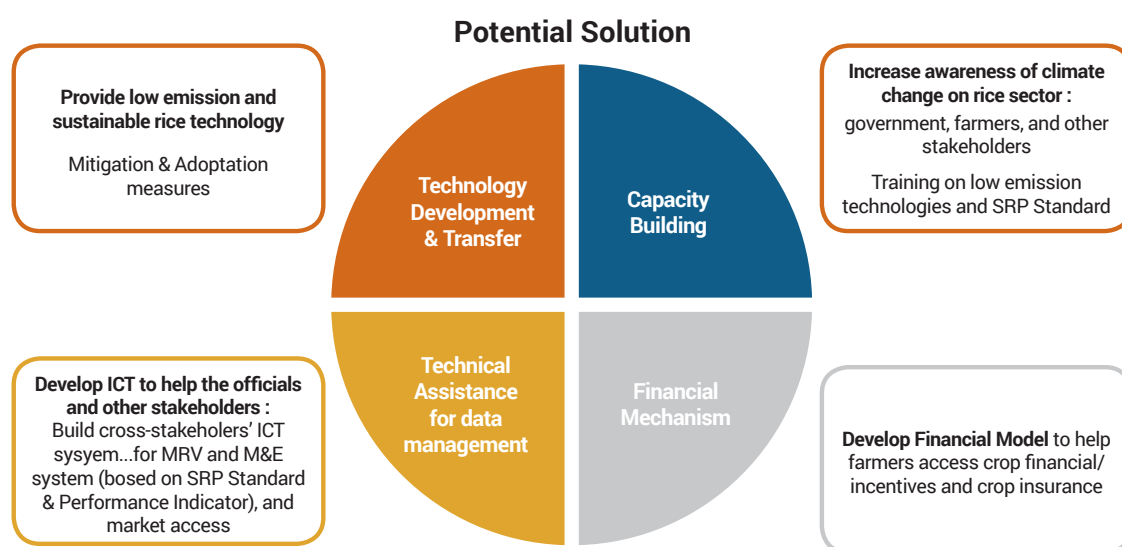
Source: Author's analysis.

CHAPTER
5.ALTERNATIVE MITIGATION
PACKAGES

Based on the Third Biennial Report (TBUR) published by ONEP in December 2020, the Thai government would like to focus on adaptation for the agriculture sector. Adaptation measures aim to improve climate resilience in this agriculture sector, such as varietal improvement, agronomic management, and water management. With the experience of extensive drought in the past years, an early warning system of extreme weather changes would also be needed. Sustainable agricultural value chains can also increase cost-effectiveness and climate resilience. Thus, in this paper, the researcher would like to suggest combining the low emission practices with other adaptation measures that could be more attractive to the farmers and for the Thai government.

Based on this desk analysis, there are four areas that could be tackled by the CCAC project:

Figure 12. Potential Solutions to support methane reduction in the Thai rice sector



The four areas shown in Figure 12 are adapted from ONEP's assessment of the TBUR 2020. These are four areas where ONEP identified needed inputs from the international agencies, which are: 1) improvement of technology development and transfer; 2) technical support / assistance to support the transfer of technical data, know-how and knowledge, including MRV and M&E System; 3) capacity building; and 4) financial support.

5.1 Technology Development and Transfer

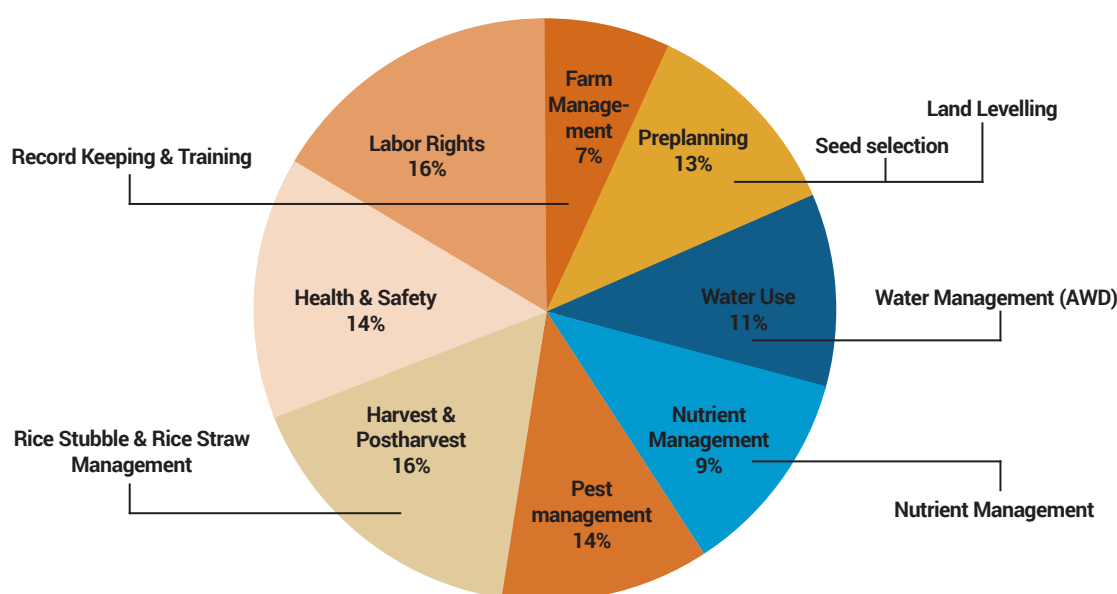
A combination of various rice technologies could achieve effective mitigation actions in the Thai rice sector. In chapter 2 and chapter 5, the researcher has described various technologies that have been widely tested and various challenges from other projects to implement the low emission technology projects. From the findings and the discussions with several stakeholders, introducing the mitigation package technologies in combination with adaptation solutions as a package would be recommended. Considering the adoption of SRP standard in Thailand that has been conducted in the past years, the consultant suggests the project scale up the

adoption of SRP Standard in Thailand under this project, promoting the SRP Standard as the Thai Rice GAP++.

1. SRP Standard and System

Adoption of the SRP Standard for Sustainable Rice Cultivation has proved successful in Thailand as well as elsewhere. As of 2020, GIZ reports almost 140,000 farmers in Thailand have been trained in the requirements of the SRP Standard and have adopted the recommended best practices. The trained farmers proved to have better capacity in managing the field, and GIZ has reported cost savings of up to 40%. As described in the previous chapter, the SRP Standard v 2.1 covers 41 requirements that comprehensively combine the climate change mitigation and adaptation measures. Figure 13 below identifies some requirements advocated under the SRP Standard that can contribute directly to GHG mitigation in rice fields.

Figure 13. Best practice mitigation measures advocated under the SRP Standard v 2.1



By emphasizing these proven best practices, the project could enhance mitigation benefits particularly for irrigated rice farms in Thailand. When combined with other low-emission technologies, the benefit would be further increased. It is recommended that relevant government agencies focus on further scale-up of adoption of the SRP Standard by farmers, support improvement of the process for assessment of mitigation impacts and support the upgrading of value chains incorporating the SRP standard for the domestic as well as for export markets.

a) Farm Management

Although **record keeping** does not at first glance seem related to mitigation, good records support the estimation of emissions, which tracks the mitigation impact. Record keeping could be adjusted from one project to another, depending on the indicator to be monitored by the project, and which assurance system is used in the project. Other than measuring the mitigation impact, having a record would help farmers see their progress and encourage them to adopt the best practices.

Training and capacity building on the SRP Standard are needed for the farmers to adopt the best practices. Training should not only include farming techniques but also the knowledge of climate change impacts and the importance of climate change mitigation and adaptation measures.

b) Pre-planting

On the SRP standard requirement number nine, it is suggested that farmers use **pure quality seeds** or certified seeds. The varieties should also be locally adapted. TRD is developing a seed variety that has shorter growing period, which would need less water. This would help farmers reduce the methane emissions due to the shorter period of planting. There can be some barriers to adoption of new varieties: farmers may face difficulties to market the new varieties if they have different tastes and characteristics that are not preferred by the market. The project could support the government's efforts to promote this new seed variety, once it is developed, tested, and approved.

Another technology under pre-planting is the Land Levelling. The Rice NAMA project promoted the **Land Laser Levelling (LLL)** for their project farmers, helping them increase the resilience of the crops. IRRI noted that LLL could improve rice yield by 24% in one of their studies in Cambodia. It is recommended to level the land with LLL prior to the implementation of AWD, to decrease the drying of high points in the field. Lastly, LLL would help the crops to effectively absorb the nutrient of fertiliser due to an even high point on the field. Thai Rice NAMA project is currently still working on this analysis, and the result could be further assessed by the project for its future project planning. It is recommended that the project to scale-up the approach that has been done by Thai Rice NAMA, especially in the distribution and the promotion of LLL technology and machineries.

c) Water Use

The **Alternate Wetting and Drying (AWD)** irrigation practice has been suggested as a water management technique to effectively mitigate methane emissions in paddy fields (LaHue et al., 2016; Liang et al., 2016; Pandey et al., 2014). In the Thai Rice NAMA, AWD was implemented in combination with Land Laser Levelling and the site-specific fertiliser application. However, it is important to remember that AWD can only be practiced on irrigated lands, and only 25% of rice in Thailand is grown under irrigated conditions. Therefore, AWD is better suited to the central provinces of Thailand.

Meanwhile, IRRI also conducted a study to estimate that AWD scenario in the six central provinces of Thai Rice NAMA could reduce annual methane emissions by 32% compared to the continuous flooded scenario, which is equivalent to 57,600t CH₄ per year, equivalent to 1.61 Mt CO₂ eq. per year (Prangbang et al., 2020). The study found that these central provinces are all largely suitable to the AWD technique in the wet season and to a large extent the dry season as well.

The IPCC 2006 guidelines and 2019 refinement provide a methodology for estimating the methane emissions from rice. This method requires data on area of rice under each water regime. The IPCC methods provide a "default" emission factor (Tier 1 method), but a country-specific emission factor (Tier 2 method) provides higher accuracy.⁷

To estimate the methane emissions from rice, the IPCC provides methodological guidance. The following is from IPCC 2019 methods refinement, using Tier 1 and Tier 2 approaches to calculate the methane emission on the rice sector, where the scaling factor for soil type and rice cultivar are only applied if available under the Tier 2 method.

$$EF = (EF_c \times SF_w \times SF_p \times SF_o \times SF_s \times SF_r)$$

Whereas:

- EF is the adjusted daily emissions factor for a particular harvested area
- EF_c is the baseline emissions factor for continuously flooded fields without organic amendments

⁷ IPCC methods can be find here: <https://www.ipcc-nggip.iges.or.jp/public/index.html>

- SF_w is the scaling factor to account for the differences in water regime during the cultivation period
- SF_p is the scaling factor to account for the differences in water regime in the season before the cultivation period
- SF_o is the scaling factor that accounts for differences in both type and amount of organic amendment applied source
- SF_s is the scaling factor for soil type, if available
- SF_r is the scaling factor for rice cultivar, if available

These IPCC methods were developed for countries to estimate their national GHG emissions but can also be applied on a project-level basis.

d) Nutrient Management

Site Specific Nutrient Management (SSNM)

Having a soil test at the beginning of the crop season would help farmers to not saturate the soil with the fertiliser. In 2005 SSNM was introduced, however only around 50% of farmers who has been introduced to this technology were implementing this practice. Attanandana et al. (2010) showed that the farmers who implemented the SSNM technique could reduce the input cost to up to 40%, meanwhile fertiliser cost would amount to around 51% of the rice production cost. Moreover, overuse of fertiliser triggers the production of N_2O . SSNM could lower the Global Warming Potential (GWP) by about 2.5% in rice (Sapkota et al., 2021). Further research to justify this calculation in Thailand could be later done by the project.

e) Harvest and Post-harvest

Stubble and Straw Management

Based on the interview with the Thai Rice Department (TRD), there is a commitment from the government for a new policy to stop straw burning on rice fields. Although it would not lead to significant reduction of greenhouse gas emission on rice production, it would reduce the air pollution significantly. Thailand has suffered from the air pollution which can reach the PM 2.5 level, esp. during the dry season (Chapter 2). Thus, a more serious effort to reduce open burning, and finding solutions to help farmers collect and gain from the rice residues should be further explored.

Learning from the Thai Rice NAMA project in the central plain of Thailand, the project helps farmers collect straw with the straw baler machine and find buyers for the straw. The project could also extend the potential to work with straw bales buyers and produce rice biproduct from rice straw for biogas, or to be further processed for furniture material. Meanwhile, seen from the MSVC project in the Northeast of Thailand, the straw and stubble are mostly left in the field, since the crop cycle is only done once a year.

2. Incentives to adopt best practices under the SRP Standard

The SRP Standard is a global standard that promotes sustainable rice cultivation. Unlike other sustainability standards, the SRP Standard is complemented by a set of **Performance Indicators (PI)** to help users measure and monitor the sustainability impact. There are 12 indicators that are defined by the SRP members as sustainability parameter to help the rice sector to be more sustainable (Fig 13). In addition, SRP also established **SRP Assurance Scheme**, where it covers the guidelines to assure the compliance of SRP standard on the field. It would help other stakeholder mainly the buyers (millers, trader, until the consumers) to ensure that the paddy production meeting the SRP Standard. It allows to include how to verify the production system, and how to communicate it to the market. To help the adopting countries or projects globally, SRP also allows each participating country to establish a **National Chapter** and adopt the standard according to the local condition.

From the available tools, the SRP Standard could be flexibly adopted in the project, depending on the available resources and the market demand. In regards of the upcoming project, the use of the tools could be optimized by the following approaches:

- a) **Using SRP the Performance Indicators for monitoring mitigation and adaptation measures**
From the communication with GIZ, the BRIA project has observed that farmers could improve their income by 40% when the SRP Standard was adopted. The Standard would help farmers improve their farming management and use the input more effectively. The effective farming would improve productivity and decrease production cost at the same time. Moreover, based on results from IRRI, the improved water management practices would reduce GHG emission up to 50% and save water use up to 20% (SRP Secretariat, 2020). The project could utilize the Performance Indicator methodology and approach for its monitoring tools and focuses on the emission mitigation parameters that is reflected in Figure 14, in red boxes.

Figure 14. SRP Performance Indicators with indicators related to mitigation in red boxes



- b) **Access to guaranteed rice market**

When talking about sustainability standards, many people assume that price incentives or premiums would be given. However, SRP has another approach, as discussed in the previous chapter. The flexibility of the assurance scheme should be further explored to build a sustainable market locally. Based on the discussion with the TRD, it is found that the current GAP standard that is promoted by the TRD has been broadly adopted in Thailand, and the farmers have seen improvement on product quality. However, the increase of product quality has not been supported with a fair pricing system and there was a lack of standard quality valuation from the miller and trader. With the adoption of the SRP Standard system in the improved GAP++ standard, it is expected that the product quality system assessment would be improved and made more adoptable by the rice stakeholders in Thailand. It is also expected that market access would improve, and the product would be valued with a fairer price based on its quality. Awareness raising on the sustainable rice product and its quality should be endorsed by the project. It is also important to communicate the product with a familiar term, such as a GAP++ product, rather than introducing the product with a new term, such as SRP rice.

Another market opportunity is to tap on the green market, which could allow “green” rice to get a tariff discount when exported. This option would need further research and exploration.

- c) **Integration with a national Standard**

The agriculture sector in Thailand has its own standard called Thai Good Agriculture Prac-

tices (GAP) standard. The SRP National Chapter Guidelines allow the implementing country to build a local chapter to adjust the SRP framework to be more relevant in each country. The National Chapter has been established in Thailand. It is coordinated by GIZ and hosted by the Thai Rice Department. The SRP National Chapter in Thailand has envisioned becoming the premier forum for driving sustainability in the Thai rice sector. The national chapter is hosted by GIZ and chaired by the Thai Rice Department, with nearly twenty stakeholder members.

In the last quarter of 2020, the chapter has committed to their goals until December 2021:

- 1) To engage the Ministry of Commerce, Thai Rice Exporters Association and Thai Rice Mills Association, and the Farmer Parliament
- 2) To develop SRP training curriculum and material in Thai Language
- 3) To develop key messages and communication materials on SRP / GAP ++

The National Chapter would develop the National Interpretation Guidelines to ensure that the SRP Global Standard is applicable in Thailand. The National Guidelines would be communicated as GAP ++, which is the upgraded version of Thai Rice GAP that has been introduced and implemented in Thailand for the past 10 years. By improving the GAP into GAP ++, Thailand would advance its national standard and would open more market opportunities globally.

5.2 Capacity Building

Almost 140,000 small-holder farmers have been trained and implemented SRP Standard in Thailand. All these projects have trained the farmers with the SRP Standard Requirements. However, the capacity building should see beyond farming technology and standards training only, it should include raising awareness on sustainable rice farming, and on climate change, and to understand the importance of the adaptation / mitigation measures.

Apart from the capacity building for the farmers, other stakeholders along the value chain should also be capacitated with the same awareness to sustain the approach. Another critical stakeholder is the officials at the bureau level and on the field level (extension workers) who would carry the scaling up of the approach in the long term. Moreover, other ministries would also need to be capacitated, especially the ministry who works closely on the climate topics, such as the departments under MoNRE (e.g., ONEP and PCD)

The capacity building effort could cover various topics such as:

1. Awareness of Climate Change
2. Mitigation and Adaptation Technologies in Rice Sector
3. SRP standard and framework
4. Farm Management and access to market
5. Finance Access
6. ICT Platform

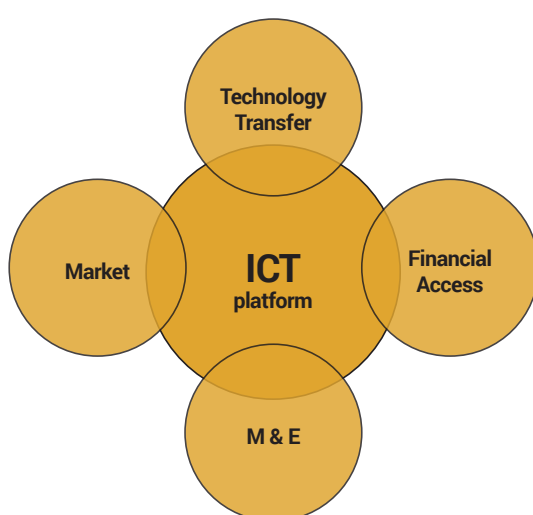
5.3 Integrated ICT Platform for Thailand

Answering the concern raised in the TBUR (2020) of Thailand, development of ICT tools could be one of the integrated solutions that could cover knowledge sharing, data collection, and the impact calculation of sustainable rice farming in a single platform. The ICT could be built to link various stakeholders along the value chain and developed within an easy and practical feature that could be adjusted for the stakeholders. Taking the pandemic into consideration, travelling and face-to-face meeting would be more difficult in the future.

Farmers could benefit from an integrated platform to receive technical advice training, and other information related to weather forecast, input price, and selling price. On the other hand, the government, to an extent probably the millers and traders as well, could use a tool to help them conduct the M&E, such as for the MRV (measurement, reporting and verification) systems for rice production that can provide ex-ante assessments as well as ex-post assessments for documenting project impacts. GHG assessments for rice are typically based on the IPCC guidelines that can be applied in various forms including with the use of GHG calculators. With such limited ICT in the rice sector in Thailand, there's a huge opportunity for the project to explore and develop an integrated system that could link the farm data to the mitigation calculation and market access.

The features of this ICT platform should be developed based on the area of support that mentioned by ONEP on the TBUR 2020 (Figure 15).

Figure 15. Proposed ideas of Features to be considered for the ICT Platform development



a) **Technology Transfer**

The technology transfer would see how information exchange between stakeholders can be supported. The main target user for this feature would be the farmers, who will receive training, and technical information or advice from experts.

There could also be an integration with Crop Manager tools that have been developed by IRRI, to show the suggested cropping calendar. Moreover, as requested by the government, development of a climate early warning system could be included in this feature. Farmers could know better about the climate risk and the weather forecast.

b) **Monitoring and Evaluation (M&E)**

MRV System

Based on the input from the TGCP project, Thai government has developed a software used in the inventory of emissions in Thailand. The agriculture sector including the rice department, is also committed to this data collection. The system could be linked to this platform, to facilitate data collection and input to the existing MRV/inventory system.

SRP Performance Indicator (PI)

Considering the SRP PI, the project could integrate the indicators to be monitored in the project. The SRP PI has a calculation method that has been tested and developed by IRRI, which should be reliable and easy to adopt. Specifically on the calculation of the climate mitigation indicators, IRRI has developed tested tool, called SECTOR tool. It would be great if the project could use SECTOR tool as the basis.

c) **Financial Access**

The platform could be used to link stakeholders in applying the Financial Instruments that would be provided by the project. At the same time, the M&E data would be a good basis to assess the farmers' risks as potential receivers of financial support.

d) Market Access

With an integrated data management, there would be more transparency among stakeholders, including between farmers and millers / traders. The flow of information regarding market price, and indicators that could affect buying price could be accessed in this platform. Rice selling price in Thailand tends to fluctuate. Therefore, having an overview of the price could help farmers to choose when to sell their products.

5.4 Financial Access

As mentioned in the earlier chapter, BAAC is the main financial provider that disburses government support on the agriculture sector in Thailand. Although various supports have been provided, such as loans, price guarantee system and crop insurance, the programmes have not fully supported farmers to implement the climate change mitigation actions, with financial support from the government.

There are two factors that put farmers at risk, namely climate risks and market dynamics (price and effect on international trade). The existing financial instruments in Thailand mainly support farmers in adapting to climate change, and there is a lack of incentive or support provided for mitigation which is crucial to reduce carbon emission from rice cultivation.

From other projects and research, the following are the recommendations of various potential financial and market instruments that were explored to address this challenge.

1. Green Loan

With the verification system of SRP, farmers who adopts SRP standard could access the funding. The loan that is provided would be given for 3-5 years and can be granted to both individuals and groups. The loan can be used as capital for the farmers to invest on farming. The loan could be used for rice farming needs such as seeding machinery, LLL services, rice seeds, and fertiliser.

Moreover, with the SRP scorecard and indicators that are integrated to the ICT, the farmers could access the loan more easily due to the more straightforward administrative process and to get easier access in accessing the loan, due to the risk reduction from the monitoring system on the ICT. This green loan scheme is suitable for individual farmers or groups of farmers/cooperatives. Collective green loan will help support farmers who do not have land tenure.

2. Improved Crop Insurance – integrated with SRP system

BAAC has effectively covered 92% of target area for rice crop insurance in 2018, however with severe droughts during the past few years, many farmers still have not been covered. One of the issues to claim the insurance is data quality, and lack of understanding from farmers on the insurance product. Moreover, crop insurance in Thailand mainly focuses on protection against natural disasters, pests, and diseases. Crop insurance is a key instrument for farmers to manage financial risks of a changing climate.

With the long and well-established crop insurance system in Thailand, it could be compatible to integrate SRP parameters into crop insurance schemes as an additional tool to measure rice producers' level of compliance and track their performance progress against all adaptation and mitigation indicators. The integrated ICT system is expected to ease the claim process with simpler procedure for farmers and this would also help BAAC in verifying the data. The insurance should also cover climate challenges not only catastrophic events, but also on peak- and off-season of rice.

3. Buyers guarantee contract / off-take agreement

Off-take agreement means a company makes a promise to buy a certain volume of a crop at a fixed price at a certain date in the future. This agreement can serve as an incentive for farmers to produce rice sustainably and it provides an assurance to the farmers that their rice produce will be purchased by committed buyers at a fixed date and price. This instrument is particularly helpful for smallholder farmers who are at risk of not being able to sell their products. Off-take agreement will mitigate the risk, and it can create three-way relationships between farmers, processors, and banks where banks provide credit to farmers, farmers sell their produce on a forward contract to processors, and processors then repay the loan to the banks (USAID, 2016).

The success of GIZ's work on MSVC project with Olam and the SARI project with Mars Food could be replicated and scaled-up in another region. Both projects used the SRP Standard verification to ensure the quality of the rice for their supply chain. Even the rice is not sold in segregated sustainable rice market, however, from the SRP standard adoption, the quality of rice has improved. It also helps the buyer to monitor the rice quality parameter. The verification model follows the SRP assurance scheme guideline levels 1 and 2. For these types of agreements, it is important that they are implemented in a socially responsible way which does not take advantage of either party in the agreement. Lessons from UNEP's Climate Smart Rice project in Myanmar could be applied here, where a socially responsible contract template was developed.

4. Multi-donor Fund

Multi-donor fund is a type of funding mechanism which involves several donors to work towards the same financing objective. Multi-donor funds are typically managed by development banks, such as the World Bank, which manages multi-donor funds specifically for SDG projects implementation, or it can also be managed jointly by the donors themselves with an establishment of a joint secretariat.

One of the examples of this model is the Thai Rice NAMA program. This program is funded by NAMA Facility but also working on two funding mechanisms (ESG, 2019):

- a. **Revolving Fund**, which is intended to support farmers who want to convert into low-emission production methods. The fund (in liquidity/cash) will be used to pay service providers upfront, and farmers will pay in instalment.
Some examples of the utilization of the revolving fund include: Laser Land Levelling (LLL), the management of straw and other harvest by-products, alternate paddy-field wetting and drying (AWD), and management of fertilizer application.
- b. **Green Loans** – subsidized by BAAC for service providers, e.g., companies offering laser and land-levelling services.

Thai Rice NAMA is an ongoing programme and there is a strong potential to scale up the projects and make a bigger mitigation impact across six provinces in Thailand. The Facility expects to generate additional 21.5 million EUR direct financial investments from the private sector and the Thai Government earmarks at least another 25 million EUR annually in agriculture- and mitigation-related areas (NAMA Facility, 2021).

Since most of the financial models in Thailand are mainly sourced from the government subsidy, and disbursed from government subsidy, it might be good to also combine the funding with other sources such as: international organisations and the private sector (e.g., private banks).

5. Carbon Credit

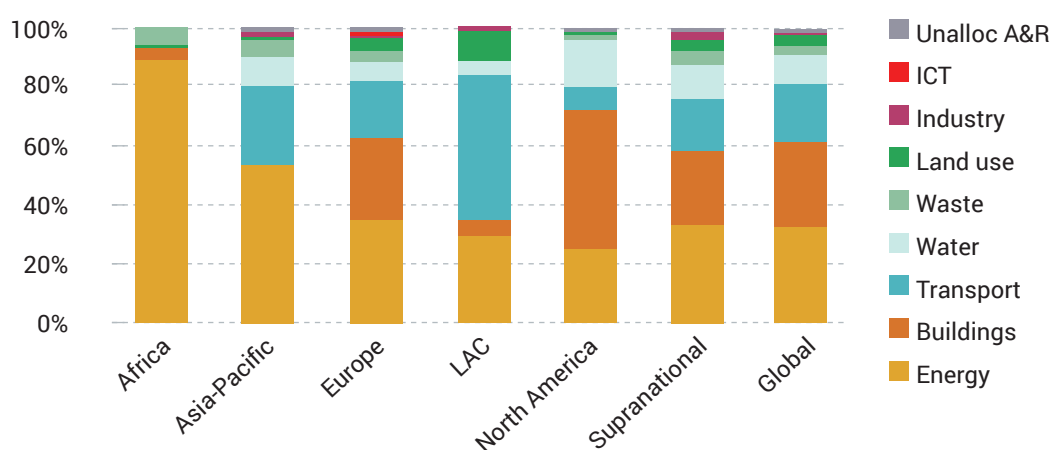
Carbon trade is one of the market mechanism instruments under Article 6 of the Paris Agreement, and it is an option for meeting the mitigation needs at a lower cost. Sustainable Rice Alliance in the US has set an example of how rice farmers can reduce methane emissions from rice cultivation and turned the reduction into carbon credit to be sold in the carbon market (Microsoft, May 10, 2018). The farmers adopted the AWD method, which in the US case reduced up to 60-90% of the emission from the business-as-usual level. This was the first ever carbon credit generated from sustainable rice cultivation.

For companies, purchasing carbon credit is a way to offset emissions generated from their activities. Carbon markets have a great potential to reduce emissions.

6. Green Bonds for Sustainable Rice

Green bonds are one of the financing options available for private firms and public entities to support climate and environmental investments. The specific use of the funds raised—to support the financing of specific projects—distinguishes green bonds from regular bonds (World Bank, 2015). At the global level, multilateral development banks have started issuing green bonds since 2007 and in 2014 a group of banks developed the Green Bond Principles (GBP)—a set of voluntary guidelines framing the issuance of green bonds. Since the establishment of the GBP, green bonds have been growing and not only banks that issue them but also other entities such as: cities, states, state-owned entities, bilateral development agencies, utilities, corporates, and national banks. In 2020 green bonds climbed to a record high of 269.5 billion USD from 266.5 billion USD the previous year. The largest issuers of green bonds by far are the United States, Germany, France, China, and Netherlands (CIO, February 2, 2021). Figure 16 shows different trends of the Use of Proceeds (UoP) in various sectors globally and in different regions. UoP reflects the allocation of funds from green bonds into projects in different sectors.

Figure 16. Use of Proceeds (UoP) Allocation in 2019



Source: Climate Bonds Initiative, 2020.

https://www.climatebonds.net/files/reports/2019_annual_highlights-final.pdf

So far, there is still a limited amount of green bonds allocated for agriculture or land use sectors. At the global level, Climate Bonds Initiative released in August 2020 an initial taxonomy for green bonds specifically for the agricultural sector (Natixis, September 24, 2020). The proposed global criteria are broad and there is ongoing work. There is a great potential to develop the scheme further, and SRP Framework can be well-integrated into the scheme.

Thailand has started issuing green bonds since 2018. Thai commercial bank, TMB, issued the

first green bond with the guidance and support from IFC as the sole investor. The proceeds raised from the 60 million USD (approximately 1,850 million THB) green bond, with a 7-year tenor, will be used exclusively to finance climate-smart projects, namely renewable energy, energy efficiency, and waste reduction (TMB Bank, 2018). Following this, a power company called B Grimm in the same year, issued a certified climate bond with the support of Asian Development Bank. This green bond is aimed at raising the share of renewable energy generation and reducing GHG emissions.

There is a great potential to develop the scheme further, and the SRP Standard (or the upcoming GAP++) and SRP Performance Indicators can be well-integrated into the scheme, particularly for M3 criteria. If SRP Framework is adopted for these criteria, it will serve as the framework for companies, banks, and investors to use as a common basis for bond design, issuance and reporting.

In Thailand, BAAC established Green Bond Framework along with ASEAN (Association of Southeast Asian Nations) Green Bonds Standard and International Capital Market Association Green Bonds Principle in August 2020 and began issuing green bonds in September 2020. BAAC is supported by Thai Security and Exchange Commission or SEC (to establish a green bond bootcamp) with the UK embassy under ASEAN Low Carbon Energy Programme, which provides external review of green bonds. Since the green bond is forestry related, stakeholders from the Ministry of Natural Resources and Environment (Biodiversity-based Economic Development / BEDO), Ministry of Interior, and Thai GHG Management Organization are also involved in the issuance of green bonds.

The proceeds will be used for two categories, namely 1) **forestry category** which aims at afforestation and reforestation projects which include: plot management and water system management that complies with forest laws; and 2) **environmental conservation category** which aims at projects that support environmental enhancement that complies with the environmental law and qualified for certain standards or certification or guidelines, such as: Good Agricultural Practice (GAP), Organic Agriculture, Participatory Guarantee System (PGS), Organic Thailand, USDA Organic, IFOAM, Good Hygiene Practice (GHP), Good Manufacturing Practice (GMP), Hazard Analysis Critical Control Point (HACCP), ISO 14001, ISO 50001 (BAAC, 2020).

Realising that GAP has been qualified for BAAC's green bonds, further consideration to include Thai Rice GAP++ with a specific aim for GHG emission reduction from the rice sector can be further explored. The proceeds from green bond or rice bond can help address the need for upfront capital for companies, suppliers, and farmers from conventional rice production to switch to sustainable production. It would also enable a rice processor, trader, or retailer to provide farmers in their value chain with capital to transition to sustainable production: improve farming practices, increase yields and revenue, and become more resilient to climate risks (Earth Security Group, 2019).

5.5. Potential Support from other Thai Rice Stakeholders

Due to the covid-19 pandemic, this paper was conducted with desk study and some interviews with the related stakeholders. From the list of stakeholders in Chapter 3.3., the researcher contacted some of them and conducted interviews to get better understanding about their interests and potentials to work in the rice climate change mitigation topic further. Interviews were made with 15 stakeholder actors in Thailand who are working on the topic of Rice and Climate Change in Thailand.

A scoping committee also was formed in Thailand, engaging the key partners who aim to scale

up the climate change mitigation approach in Thailand. Although the committee was only requested to support the project for the review of this scoping study, there is a long-term intention to work with these committee members. The committee consists of four stakeholder actors, namely from the Thai Rice Department, IRRI, Herba (Ebro), SRP Secretariat and hosted by UNEP.

Table 8. Potential Contribution from various Stakeholders

Actors	Interests	Potential Contribution
Thai Rice Department	Despite of limited institutional commitment on the climate mitigation goal on rice sector, TRD keeps on working to be innovative and promoting rice farming technologies that could potentially reduce the GHG emission from rice sector - To encourage farmers to adopt good agricultural practices: less chemical fertiliser, produce better rice quality, efficient use of water - TRD is developing the short-term rice variety, which would need less water - Integrated this objective to Big rice farming project - Implementing digital technology and precision farming - Rice seed production to support higher quality rice production with lower seed rate - Support with conditional incentive	TRD as the main bureau that work on Rice related programs and policy is the key partner to advise on the project strategy, and as the project implementing partner on the field to test new approaches or solutions.
ONEP	Responsible for climate change mitigation actions in Thailand to reach the NDC of Thailand.	Advisory support from ONEP would be very much appreciated, especially in the alignment with other CCAC projects in Thailand and the NDC strategy and policy
PCD	Responsible for policy development, planning, standards setting, intra-agency and international cooperation, audit, and evaluation of pollution-related issues.	As the focal point for the CCAC program in Thailand, working with PCD would be very beneficial to align the program and to ensure that the project could contribute to the CCAC programme impact in Thailand.
Farmers through CRC	To gain more yield and profit, to be able to have a constant crop cycle and not to worry about climate events	Collaboration in the project and help to organize the farmers
IRRI	To advance and sustain rice farming technology. To encourage the use of the ICT / calculation tools	As IRRI's experts have been working on this topic before, technical inputs to foster the sustainable and climate friendly rice cultivation could be given. Moreover, the available ICT tools (Farm Management, and SECTOR tool) could be integrated to the planned ICT system for this project.
Herba Thailand (Ebro)	To transform the business into full sustainable rice sourcing by 2030 and to reach the company SDG goals	To test the market linkages model for sustainable paddy, both for local and international markets. Moreover, to also promote the products worldwide (potentially create an international sustainable rice market)
Olam	To distribute good quality, safety and low emitted rice	

UNEP	To help Thailand reach its NDC target and the goal and mission of the CCAC program in Thailand	To host the project, and support the project implementation with its expertise and infrastructure in Thailand
SRP Secretariat	SRP Secretariat would like to see broad adoption of the SRP framework and to support the farmers to gain more income, and to improve the health and safety of the rice paddies.	Provide advisory support on how best SRP framework and tools could be used to reach the NDC
SRP National Chapter	On the last quartal of 2020, the chapter has committed to their goals until December 2021: 1) To engage the Ministry of Commerce, Thai Rice Exporters Association and Thai Rice Mills Association, and the Farmer Parliament 2) To develop SRP training curriculum and material in Thai Language 3) To develop key messages and communication materials on SRP / GAP ++	Become a platform where the project could engage and broaden its network with the SRP stakeholders in Thailand
GIZ	<i>Depends on the project</i> To support the beneficiaries' countries in reaching its NDC and to improve sustainable rice production – improve farmers' income	GIZ has been working in the Thai Rice sector for the past 7 years and has been equipped with a strong infrastructure on the field level. GIZ has also established good cooperation with other stakeholders (policy makers, trader, millers). Collaborating with GIZ as a technical implementing partner would be beneficial for the project
Kasetsart University & Khon Kaen University	They are familiar with the topic and would like to explore opportunities to conduct the study with the support of the projects (financially)	Both universities have a broad network in Thailand and could support the project to conduct field study or research

From the assessment above, there is potential to scale-up the technologies and approach a collaboration with various stakeholders. Further stakeholder assessments should be done once a clearer approach and direction is determined by the project. During the study, the researcher faced difficulties finding a local NGO who has been working on this topic. Considering that UNEP/CCAC does not have a strong infrastructure on the ground, further consideration to collaborate with local NGO or institutions would be very valuable. It would help the project outreach and help maintain a good partnership with the other value chain actors in Thailand.

CHAPTER

6.

CONCLUSIONS and
RECOMMENDATIONS

This scoping study has aimed to explore various potentials and available resources that could be further used by the project to upscale the mitigation technology in Thailand. Based on the existing projects, there is still a huge potential for the project to make greater impact and to scale-up the current approaches. Limited awareness from various stakeholders on the issue could be seen as a potential to transform the value chain to be more sustainable and more climate friendly.

Moreover, with the adjusted NDC of Thailand, there is actually a big potential for Thai government to include the rice sector into their mitigation calculation. On the other hand, the project should be aware of introducing the mitigation technology. Considering the farmers condition and the interest of the government, the mitigation package should be integrated with some adaptation solutions to be more relevant and attractive for the beneficiaries. Joint collaboration among various stakeholders including the policy maker would be the key to success in this project implementation. Although the approaches that were suggested are not novel, there is still rooms for improvement. Commitment from all related stakeholders would be crucial.

Despite the great potential, there are some challenges and barriers, especially on the production or farmers. In the past years, climate change has impacted farmers. Increased temperatures result in both drought and flood events. With Thai farmers ageing, this condition would make it more difficult for them to adopt new technologies and understand new concepts such as sustainability and climate change mitigation. To create a great impact, the project should explore ICT solutions that could fit into the average farmers age group in term of its practicality and accessibility.

UNEP would need to work closer with the MoAC and MoNRE, with other international organizations active in Thailand, and to find local implementing partners that can support communication and engagement on the ground. From the four area of potential solutions that are described in chapter 5, the improvement of climate mitigation agriculture solution and capacity building would be the most typical development agriculture solutions. However, the project could pay more attention on the Financial Mechanism development and the ICT platform development, as it is not well developed yet and accessible in Thailand. Despite a great adoption of climate mitigation actions by farmers, if there is no sufficient data system and practical monitoring tools on the field, it would be hard to measure the impact. Focuses on ICT platform development would complement the work of TRD and other ongoing projects. Considering UNEP's mandate to provide policy guidance, UNEP could also focus more on the policy dialogue and strategic solutions. It is very important to align the work with other agencies, and private institutions or start-ups company to create greater impact and would bring a new perspective into the project.

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